

Privacy Risks and Leakage Pathways in Agentic and Multi-Agent AI Systems

ARASH MAHBOUBI, School of Professional Studies, UNSW Canberra, Canberra, Australia

SEYIT CAMTEPE, Commonwealth Scientific and Industrial Research Organisation, Canberra, Australia

KEYVAN ANSARI, School of Information Technology, Murdoch University, Murdoch, Australia

THANH HANG BUI, UNSW Sydney, Canberra, Australia

HAMED ABOUTORAB, UNSW Sydney, Canberra, Australia

Agentic and multi-agent AI systems are increasingly deployed as part of operational systems, introducing autonomous planning, delegation, persistent memory, tool use, and inter-agent coordination into data-driven decision making. These architectural properties challenge prevailing privacy assumptions because leakage may arise structurally from autonomy, coordination, persistence, and composition rather than only from isolated model outputs. This paper provides a structured synthesis of privacy risks and leakage pathways for agentic and multi-agent AI systems. We introduce a unifying abstraction that combines decentralized decision making with trace-based information-flow semantics, enabling reasoning about autonomy, coordination, persistent state, observability, and adversarial trace projections. Using this abstraction, we organize recurring privacy risks into five leakage pathways: autonomy-induced overexposure, coordination-driven information propagation, memory and persistent-state leakage, emergent and compositional inference, and observability and audit leakage. We further analyze why established privacy-enhancing technologies provide incomplete coverage in agentic settings, where static computation boundaries, single-agent assumptions, ephemeral state, and non-emergent behavior often fail. To ground the analysis, we implement a controlled multi-agent privacy sandbox with privacy-aware agents, tool-mediated workflows, adversarial trace projections, and LLM-backed validation. The results show that policy-checked privacy-aware execution reduces direct disclosure and adversarial inference advantage, while residual tool-mediated and semantic-hint leakage remains. The paper concludes by proposing new privacy requirements for agentic and multi-agent AI systems.

CCS Concepts: • **Security and privacy** → **Privacy protections**; Information flow control; • **Computing methodologies** → *Multi-agent systems*; *Intelligent agents*.

Additional Key Words and Phrases: Privacy, Multi-Agent Systems, Agentic AI, Differential Privacy, Privacy-Enhancing Technologies, Healthcare

1 Introduction

Agentic AI and multi-agent systems (MAS) extend conventional machine-learning systems by supporting autonomous planning, tool use, task delegation, inter-agent communication, and adaptation over extended interactions [13]. These capabilities introduce privacy concerns that are not adequately captured by models developed for static computation pipelines. Information may propagate through inter-agent messages, shared state, persistent

Authors' Contact Information: Arash Mahboubi, School of Professional Studies, UNSW Canberra, Canberra, Australian Capital Territory, Australia; e-mail: a.mahboubi@unsw.edu.au; Seyit Camtepe, Commonwealth Scientific and Industrial Research Organisation, Canberra, Australian Capital Territory, Australia; e-mail: Seyit.Camtepe@csiro.au; Keyvan Ansari, School of Information Technology, Murdoch University, Murdoch, WA, Australia; e-mail: Keyvan.Ansari@murdoch.edu.au; Thanh Hang Bui, UNSW Sydney, Canberra, Australian Capital Territory, Australia; e-mail: hangthanhbui@unsw.edu.au; Hamed Aboutorab, UNSW Sydney, Canberra, Australian Capital Territory, Australia; e-mail: h.aboutorab@unsw.edu.au.



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memory, delegated actions, and external tools, making the source and extent of a disclosure difficult to determine. Existing privacy frameworks and privacy-enhancing technologies (PETs) generally assume well-defined computation boundaries, controlled data flows, and identifiable data controllers. These assumptions become difficult to maintain when autonomous agents coordinate across organisational and technical boundaries. Privacy leakage may therefore arise not only from individual outputs, but also from the composition of observations, actions, intermediate messages, and retained state over time.

Healthcare provides a representative setting in which these risks are particularly consequential. Agentic systems are being explored for clinical decision support, patient interaction, medication management, and operational coordination. Although individual agents may operate under role-specific access constraints, their coordination can propagate sensitive information beyond its intended context. For example, an eligibility or resource-allocation request may reveal a sensitive diagnosis indirectly through its content, timing, or downstream effects.

Existing healthcare systems already illustrate the difficulty of maintaining privacy across distributed and multi-stakeholder infrastructures. In September 2025, confidential documents relating to nearly 600 medical staff were accidentally exposed through a shared New South Wales local health district web system [20]. Agentic and multi-agent architectures add further complexity through autonomous delegation, persistent state, and cross-agent information flows, motivating a systematic analysis of their privacy risks and leakage pathways.

Recent AI-assistant and agentic-system incidents show that these risks are not merely hypothetical. Enterprise copilots and assistant-integrated workflows have exhibited privacy-relevant failure modes involving indirect prompt injection, retrieval-augmented exposure, tool-mediated exfiltration, and cross-context information propagation. EchoLeak/CVE-2025-32711 in Microsoft 365 Copilot demonstrated how a zero-click prompt-injection chain could turn an assistant with access to organisational context into a channel for unauthorised information disclosure [31]. Related work by J. Rehberger showed that Microsoft 365 Copilot could be manipulated through prompt injection and tool invocation to exfiltrate personal information such as emails and contextual user data [33]. Similar risks have also been reported for collaboration platforms: the Slack AI prompt-injection vulnerability disclosed by PromptArmor showed how private-channel content could be exposed through AI-assisted search and summarisation workflows [23]. In developer environments, GitHub Copilot and VS Code attack chains have further shown that assistant-mediated tool use can contribute to confidential-file exposure, token leakage, or remote-code-execution risk when prompt injection reaches agentic coding workflows [34].

These observations point to two intertwined challenges. First, autonomy and delegated decision-making obscure accountability boundaries, complicating the attribution of responsibility for privacy violations. Second, the composition of agent behaviours gives rise to leakage pathways that evade traditional access control, anonymisation, and logging mechanisms. Established PETs such as differential privacy, secure multiparty computation, and federated learning offer important protections in conventional settings, but their guarantees rely on assumptions—such as static data flows, single-agent reasoning, and ephemeral state—that are routinely violated in agentic and multi-agent environments. In particular, these technologies are not designed to address privacy risks arising from persistent memory, inter-agent coordination, chain-of-thought sharing, or task delegation across heterogeneous trust domains.

This paper argues that agentic and multi-agent AI systems constitute a distinct and increasingly important frontier for privacy research. Prior work has examined privacy in machine learning and federated settings, security and safety concerns in autonomous agents [14, 16], contextual integrity in information-flow privacy [3, 25], and recent benchmarks for privacy norm awareness and leakage in (Large Language Model) LLM-based and tool-mediated agent systems [9, 17, 21, 37]. However, these lines of work remain fragmented across privacy theory, benchmark design, agent safety, distributed AI, and privacy-enhancing technologies. We therefore provide a structured systematization that connects these strands through a formal, architecture-agnostic abstraction for reasoning about autonomy, delegation, inter-agent communication, persistent state, adversarial observability,

and emergent privacy leakage. Based on this analysis, we identify structural sources of privacy risk, examine why existing PETs provide limited protection in these settings, and discuss implications for the design of privacy-preserving agentic AI systems.

Controlled empirical grounding of agentic and MAS privacy risks. We introduce a multi-agent privacy sandbox that instantiates the formal trace model under controlled clinical workflows. The revised sandbox includes privacy-aware agents, explicit policy-checking modes, tool-mediated workflows, adversarial trace projections, and LLM-backed validation through a local instruction-following model. The sandbox is not presented as a replacement for public benchmarks, a leaderboard, or evidence that privacy is solved. Rather, it provides controlled empirical calibration of how direct disclosure, semantic hints, tool-mediated leakage, residual structural leakage, and adversarial inference advantage arise under different privacy modes and architectures.

Recent public-sector AI initiatives emphasise autonomous task execution, delegated decision-making, and cross-organisational coordination [18, 36], directly motivating our analysis focus on privacy risks that arise specifically from autonomy, coordination, delegation, and inter-agent information flows in agentic and multi-agent AI systems. We do not attempt to comprehensively survey broader AI safety, robustness, or purely security-related attacks, except where these intersect directly with privacy. Our scope includes both theoretical models and emerging real-world deployments, spanning decentralised and centrally orchestrated MAS architectures across organisational and cross-organisational boundaries. This paper makes the following contributions:

- i. **A structured synthesis and formal abstraction for agentic/MAS privacy.** We synthesize emerging work on privacy in agentic and multi-agent AI and organize it through a formal hybrid model that combines decentralised partially observable Markov decision processes (Dec-POMDPs) with trace-based privacy and information-flow semantics.
- ii. **A unified taxonomy of privacy leakage pathways.** We introduce a structured taxonomy that captures privacy risks arising from autonomy, coordination, delegation, persistent memory, inference, and emergent multi-agent behaviour.
- iii. **Controlled empirical grounding through a multi-agent privacy sandbox.** We introduce a light-weight, synthetic multi-agent privacy sandbox that instantiates the formal system and threat models under controlled clinical workflows. The sandbox includes privacy-aware agents, explicit policy-checking modes, tool-mediated workflows, adversarial trace projections, and LLM-backed validation through a local instruction-following model.
- iv. **Benchmark-style calibration against related privacy-evaluation settings.** We calibrate the sandbox using scenario families inspired by contextual privacy, tool-mediated agent privacy, internal-channel leakage, multi-agent contextual privacy, and compositional leakage. These scenario families are aligned with privacy phenomena studied in ConfAide-style, PrivacyLens-style, AgentLeak-style, MAGPIE-style, and compositional-leakage evaluations.
- v. **Systematic analysis of existing PETs for agentic and MAS settings.** We analyse the applicability, guarantees, and limitations of current privacy-enhancing technologies, including differential privacy (DP), private information retrieval and query privacy, federated learning with secure aggregation, secure multi-party computation (MPC), secret sharing and threshold cryptography, trusted execution environments (TEEs), homomorphic encryption (HE), information-flow control, zero-knowledge (ZK) proofs, purpose binding, and privacy-preserving logging in the context of agentic and multi-agent architectures.
- vi. **A justified PET-to-risk mapping for agentic and MAS privacy.** We map representative privacy-enhancing technologies to the identified leakage categories and analyse where their assumptions fail under autonomy, delegation, repeated interaction, persistent memory, inter-agent coordination, and multi-agent composition. We provide cell-level rationale for the PET-to-risk ratings and discuss privacy-budget composition under repeated and correlated multi-agent outputs.

- vii. **Regulatory and governance implications for privacy-preserving agentic AI.** We discuss how agentic privacy failures interact with regulatory principles such as purpose limitation, data minimisation, consent, access control, auditability, accountability, and high-risk AI governance under frameworks such as GDPR, HIPAA, and the EU AI Act.
- viii. **Research agenda for privacy-preserving agentic AI.** We outline open challenges and future research directions for developing scalable, composable, and verifiable PETs tailored to agentic and multi-agent AI systems, including privacy-aware planning, purpose-bound delegation, memory governance, privacy-preserving observability, adversarial trace auditing, and PETs that compose across agents and over time.

2 Case Study: Agentic Multi-Agent AI in a Hospital Network

In this case study, we consider a stylised but realistic deployment of agentic, multi-agent AI in a medium to large-sized hospital network. The scenario is informed by recent industry reports describing autonomous clinical decision support, workflow orchestration, and operations optimisation in hospitals and health systems.¹ We show how the privacy challenges identified in this paper manifest in a concrete application setting, and how existing PETs (and their limitations) play out in practice. The case study also illustrates how our formal model (Dec-POMDP + trace-based information flow) can be instantiated for a realistic application.

2.1 System Overview

Consider a hospital network, *HospitalNet*, that deploys an agentic AI platform comprising three primary agents: (a) a **Clinical Decision Agent (CDA)**, which analyses electronic health records, laboratory results, vital-sign streams, and imaging summaries to suggest diagnoses and treatments, monitor clinical deterioration, and recommend interventions such as early ICU transfer; (b) a **Resource Coordination Agent (RCA)**, which manages bed allocation, operating-room scheduling, nurse staffing, and critical equipment under constraints such as staffing capacity, infection-control zoning, and emergency demand; and (c) a **Patient Engagement Agent (PEA)**, which interacts with patients and carers through chat, voice, and mobile applications to collect symptoms, answer routine questions, schedule follow-ups, and escalate safety concerns to the care team.

These agents operate over a shared technical substrate containing patient profiles, operational state, and interaction logs. Large language models fine-tuned on medical text and local documentation support the CDA and PEA, while real-time pipelines integrate EHR events, laboratory and imaging results, and streaming vital signs. An orchestration layer coordinates tasks among the CDA, RCA, PEA, and existing hospital information systems.

Such architectures are commonly motivated by anticipated improvements in clinical decision support, operational efficiency, and care coordination observed in related AI applications in healthcare. Prior research has documented that AI systems - for example, machine learning models for diagnosis and resource scheduling - can reduce diagnostic delays, reduce hospital readmissions, and improve utilisation of beds and clinical personnel when integrated into clinical workflows [29]. These prospective benefits underscore why healthcare organisations are exploring more autonomous AI solutions, even as the same design features - such as agent autonomy, inter-agent coordination, persistent shared state, and rich telemetry - may instantiate privacy risks of the kinds analysed in Section 7.

¹For example, Kodexo Labs describes agentic AI deployments spanning clinical decision support, administrative automation, and resource management in contemporary healthcare environments.

2.2 Illustrative Scenario: Agentic Multi-Agent AI in a Hospital Network

Table 1 provides a comparative overview of representative agentic AI use cases projected to influence healthcare environments in the near term. The table highlights differences in implementation complexity, resource requirements, expected clinical and operational outcomes, and typical deployment scenarios. Its purpose is to emphasise the diversity of operational settings in which the privacy risks analysed in this paper may arise.

Table 1. Contextual examples of agentic AI use cases in healthcare

AI application	Implementation complexity	Resource requirements	Expected outcomes	Ideal use cases	Key advantages
Autonomous clinical decision support systems	High; requires extensive training data, validation, and integration	High; requires computation, data access, and maintenance	Reduced diagnostic errors, continuous support, and cost efficiency	Complex diagnostics requiring multimodal data analysis	Improved accuracy, standardisation, and continuous learning
Intelligent drug discovery and development agents	High; requires diverse datasets and computational infrastructure	Very high; requires extensive computation and data	Shorter discovery timelines, reduced costs, and novel targets	Drug R&D acceleration and compound optimisation	Faster drug design, lower failure rates, and personalised medicine
Autonomous patient monitoring and care coordination	Moderate; requires IoT integration, real-time processing, and security controls	High; requires sensors, cloud services, and mobile applications	Earlier intervention, reduced readmissions, and continuity of care	Chronic disease management and remote patient care	Real-time monitoring, workload reduction, and prevention
AI-powered robotic surgery assistants	Very high; requires advanced hardware, software, and surgeon training	Very high; requires robotics infrastructure and maintenance	Enhanced precision, fewer complications, and faster recovery	Minimally invasive and high-precision procedures	Precision, consistency, and reduced human error
Intelligent medical imaging and radiology automation	Moderate; depends on system integration and training-data quality	Moderate to high; requires GPU-heavy image processing	Faster diagnosis, improved accuracy, and workload reduction	High-volume radiology departments and triage support	Speed, accuracy, and reduced clinician burden
Autonomous personalised treatment planning systems	High; requires integration of genomic and clinical data	High; requires sequencing and analytics infrastructure	Optimised therapies and reduced adverse effects	Oncology, rare diseases, and pharmacogenomics	Precision medicine and dynamic treatment adjustment
Intelligent healthcare resource management and optimisation	Moderate to high; requires complex system integration	Moderate; requires analytics engines, scheduling tools, and EHR integration	Reduced operational waste, improved staffing, and optimised workflows	Capacity planning, staff scheduling, and supply-chain optimisation	Operational efficiency, reduced bottlenecks, and cost savings

Across these diverse use cases, common architectural features—autonomy, coordination, shared state, and delegation-recur, directly motivating the structural privacy risks analysed in Section 7.

3 Background and System Model

This section introduces the foundational concepts underlying agentic and multi-agent artificial intelligence systems and presents the system abstraction used throughout this paper to analyse privacy risks. We first describe architectural characteristics of agentic and multi-agent systems that are particularly relevant to privacy. We then introduce a unifying analytical model that combines decentralised partially observable Markov decision processes (Dec-POMDPs) with trace-based semantics.

3.1 Agentic and Multi-Agent AI: Background

Agentic AI refers to artificial intelligence systems that autonomously plan, act, and interact with their environment in order to achieve high-level objectives. Such systems commonly rely on external tools, contextual memory, and dynamic reasoning mechanisms [13]. In contrast to conventional machine learning models that function as static inference components within predefined pipelines, agentic systems support goal decomposition, action sequencing, tool invocation, and online adaptation. These capabilities allow agents to operate over extended

time horizons, retain internal state, and adjust behaviour in response to evolving environmental and contextual conditions.

Recent industry and policy reports indicate rapid growth in the deployment of autonomous and adaptive AI systems within healthcare. Agentic architectures are being explored for applications such as clinical decision support, patient monitoring, care coordination, and operational optimisation. Their appeal lies in the ability to integrate heterogeneous data sources and support complex workflows [32]. These developments reflect a broader shift toward AI systems that operate with reduced direct human oversight while maintaining continuous interaction with clinical, administrative, and operational infrastructures. In subsection § 2.2 presents illustrative healthcare-oriented use cases that serve as a running example throughout this paper.

Multi-agent systems extend agentic AI by enabling multiple autonomous agents to interact, collaborate, or compete within a shared environment. Such systems may be centrally orchestrated, fully decentralised, or hybrid, and typically rely on explicit communication, shared state, and task delegation mechanisms. While multi-agent systems have long been studied in distributed artificial intelligence and control, the integration of contemporary large language model based agent frameworks substantially increases agent autonomy, memory capacity, and flexibility of coordination. As a result, new forms of emergent behaviour and information propagation arise that are especially relevant for privacy analysis.

Although healthcare scenarios are used for concreteness, the architectural properties discussed here recur across a wide range of domains, including collaborative robotics, financial decision support, and cyber defence automation. In all of these settings, increased autonomy and coordination amplify privacy risks by expanding data access, enabling persistent memory accumulation, supporting cross-agent inference, and creating complex delegation structures. Analysing how these properties interact is therefore essential for understanding privacy risks in agentic and multi-agent systems.

3.2 Decentralised Multi-Agent Decision-Making Model

We adopt the standard decentralised partially observable Markov decision process abstraction to capture autonomy and interdependent decision-making. A Dec-POMDP is defined by the tuple

$$\mathcal{M} = \langle \mathcal{A}, \mathcal{S}, \{O_i\}_{i \in \mathcal{A}}, \{A_i\}_{i \in \mathcal{A}}, P, R, O, \gamma \rangle,$$

where: (1) $\mathcal{A} = \{1, \dots, n\}$ denotes the set of agents; (2) $\mathcal{S}$ is the environment state space; (3) O_i is the observation space of agent i , with joint observation space $\mathcal{O} = \times_{i \in \mathcal{A}} O_i$; (4) A_i is the action space of agent i , with joint action space $A = \times_{i \in \mathcal{A}} A_i$; (5) $P(s' | s, \mathbf{a})$ is the state transition function; (6) $R(s, \mathbf{a})$ is the shared reward function, with individual rewards represented by $\{R_i(s, \mathbf{a})\}_{i \in \mathcal{A}}$ when required; (7) $O(o_i | s', \mathbf{a}, i)$ is the observation function for agent i ; and (8) $\gamma \in (0, 1)$ is the discount factor.

At each timestep t , the environment occupies a state s^t . Each agent receives a private observation o_i^t and selects an action based on its local history $h_i^t = (o_i^0, a_i^0, \dots, o_i^t)$. The joint action induces a state transition and a corresponding reward. A joint policy $\pi = (\pi_1, \dots, \pi_n)$ induces the joint value function

$$V^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s^t, \mathbf{a}^t) \mid s^0 = s, \pi \right].$$

The Dec-POMDP abstraction captures autonomy, partial observability, and interdependent decision-making. However, it does not explicitly represent information flow, logging, or persistent state. These aspects are essential for analysing privacy risks and are incorporated in a trace-based model and explicit representations.

3.3 Trace-Based Model for Privacy and Information Flow

We model system execution as a trace $\tau = \langle e_0, e_1, \dots, e_T \rangle$, to reason about privacy properties where each event $e_t \in \mathcal{E}$ represents an observation, action, message, or data access operation. For any entity X , such as an agent, auditor, or adversary, a projection operator $\tau \downarrow_X$ restricts the trace to the events observable by X .

A privacy requirement is formalised as a predicate $\text{Priv} : \mathcal{T} \rightarrow \{\text{true}, \text{false}\}$, where $\text{Priv}(\tau) = \text{true}$ indicates that the trace τ satisfies the required constraint. This formulation supports reasoning about data minimisation, inference, and policy compliance across heterogeneous system components without assuming a particular implementation.

3.4 Knowledge, Memory, and Shared State

Each agent maintains a private knowledge state $\kappa_i^t = (h_i^t, m_i^t, D_i)$, where h_i^t denotes the interaction history, m_i^t denotes internal memory, and D_i represents private data accessible to agent i . Persistent memory enables agents to reuse intermediate results, summaries, and contextual information across tasks and over time.

Many practical agent platforms also rely on shared state, represented by a shared knowledge base KB^t that contains system state, policies, logs, and contextual artefacts. Read and write access to KB^t enables coordination beyond classical Dec-POMDP assumptions. At the same time, it introduces additional privacy-relevant information propagation pathways that must be explicitly modelled.

3.5 System Model Summary

The system abstraction used in this paper integrates: (1) a Dec-POMDP behavioural model that captures agent autonomy and interdependent decision-making; (2) trace-based semantics for analysing information flow and privacy constraints; and (3) explicit representations of persistent agent memory and shared state.

Generalisability. This hybrid abstraction is intentionally architecture agnostic. It captures privacy-relevant properties that are shared by diverse agent implementations, including large language model based, rule based, and planning centric systems. In subsequent sections, we use this abstraction to connect architectural features such as autonomy, delegation, coordination, and persistence to recurring privacy risks. We also use it to analyse why classical privacy enhancing technologies often fail to compose effectively in agentic and multi-agent environments.

3.6 Operational Flow of the Hybrid Dec-POMDP Model

Figure 1 illustrates the operational information flow of the hybrid Dec-POMDP and trace-based privacy model. At time t , the environment is in state $s^t \in \mathcal{S}$ and evolves according to the transition function $P(s^{t+1} \mid s^t, \mathbf{a}^t)$ following the joint action $\mathbf{a}^t = (a_1^t, \dots, a_n^t)$. Each agent $i \in \mathcal{A}$ receives a private observation $o_i^t \sim O(\cdot \mid s^t, \mathbf{a}^{t-1}, i)$, reflecting decentralised control and partial observability.

Each observation updates the agent knowledge state $\kappa_i^t = (h_i^t, m_i^t, D_i)$. Based on this state, the agent selects an action according to its policy π_i , such that $a_i^t \sim \pi_i(\cdot \mid h_i^t)$. In addition to local decision making, agents may read from or write to the shared knowledge base KB^t , subject to access control rules and privacy constraints.

The environment produces a reward signal $r^t = R(s^t, \mathbf{a}^t)$, and system execution generates a trace $\tau = \langle e_0, e_1, \dots, e_T \rangle$. Each event records observations, actions, inter agent messages, or accesses to the shared knowledge base. A privacy checker evaluates whether the execution satisfies a specified privacy predicate $\text{Priv}(\tau) = \text{true}$, which may encode requirements such as data minimisation, restricted information propagation, or bounded data retention.

This operational flow makes explicit the information pathways created by autonomy, coordination, and persistence. These pathways form the basis for the threat model in Section 4, where we analyse adversarial capabilities, attack surfaces, and privacy leakage vectors that arise from these interactions.

Rationale for the Dec-POMDP Based Model. Agentic and multi-agent systems combine long horizon autonomy, decentralised control, and interdependent decision making. At present, there is no widely adopted formalism that captures these properties while also supporting reasoning about information flow and privacy. Without an explicit behavioural model, privacy risks cannot be systematically traced to specific architectural features or execution patterns.

The hybrid Dec-POMDP and trace-based abstraction introduced in this paper provides a principled foundation for analysing how autonomy, shared memory, and inter agent interaction give rise to new privacy leakage pathways. By making decision processes, observations, and information propagation explicit, the model enables structured reasoning about privacy risks before such systems are widely deployed in operational settings.

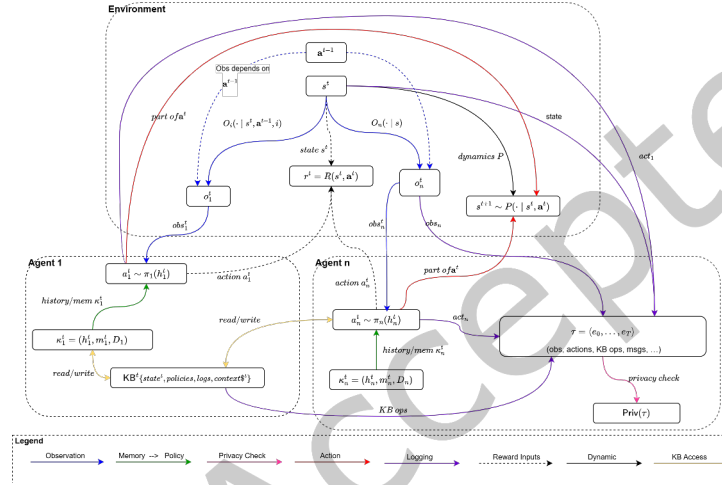


Fig. 1. Hybrid Dec-POMDP with observation $\rightarrow$ memory $\rightarrow$ policy flow. Observations at time t depend on the current state s^t and the previous joint action a^{t-1} via $O(\cdot | s^t, a^{t-1}, i)$. Colour-coded edges show dynamics, actions, KB access, logging, and privacy checking $\text{Priv}(\tau)$.

3.7 Mapping to the Formal Model

Dec-POMDP structure: The environment state $s \in \mathcal{S}$ encodes both clinical and operational information. Clinical state includes patient specific variables such as diagnoses, laboratory results, and treatment orders. Operational state includes information about beds, staff availability, medical devices, and workflow queues. We consider three principal agents, denoted by the set $\mathcal{A} = \{\text{CDA}, \text{RCA}, \text{PEA}\}$.

For each agent $i \in \mathcal{A}$:

- O_i denotes the space of local observations. For example, the clinical decision agent observes summarised electronic health records and vital signs, the resource coordination agent observes occupancy and acuity summaries, and the patient engagement agent observes patient messages and a restricted subset of clinical data.
- A_i denotes the action space. Typical actions include issuing clinical recommendations, reallocating beds or staff, and sending messages or escalation requests.
- π_i denotes the policy governing agent behaviour. Policies may be deterministic or stochastic and may be learned or manually configured.

The transition function captures how joint actions update both clinical and operational state. For instance, a clinical decision alert may lead a clinician to order additional tests, thereby modifying the clinical state and triggering downstream operational adjustments such as bed reallocation.

Trace-based information flow. System execution is modelled as a trace $\tau = \langle e_1, e_2, \dots \rangle$, where each event records agent observations, actions, state updates, or logging operations. For an entity X , such as an internal auditor, a compromised system component, or an external observer with access to selected logs, we consider the projected trace $\tau \downarrow_X$ containing only events observable by X .

A privacy policy is expressed as a predicate $\text{Priv}(\tau)$ over traces. As an illustrative example, a policy may require that no unauthorised entity is able to infer a patient sensitive diagnosis from system behaviour, including indirect signals such as changes in resource allocation or communication patterns.

In Section 4, we define classes of adversaries and their observational capabilities. In this healthcare setting, these include honest but curious operators, compromised patient engagement agents, and colluding third party service providers with partial access to system logs.

4 Threat Model

This section defines the threat model used to analyse privacy risks in agentic and multi-agent AI systems. It builds on the formal system abstraction introduced in Section 3 and specifies adversarial entities, their capabilities, their observable views of system execution, assumptions about collusion, trust boundaries, and attack surfaces. The focus is on privacy threats rather than general security attacks, except where security failures directly enable privacy leakage or undermine privacy enforcement.

4.1 System Threat Model Overview

We consider heterogeneous deployments of agentic and multi-agent systems that operate over private data, persistent agent memory, shared state, and inter-agent communication across trust boundaries. Adversaries may seek to infer, extract, propagate, or retain sensitive information beyond its intended scope or purpose. Attacks may target individual agents, the shared knowledge base KB, communication channels, logging and monitoring infrastructure, or the privacy enforcement mechanism Priv .

We distinguish several classes of adversaries, including external entities, compromised or malicious agents, honest-but-curious participants, platform operators, auditors, and third-party integrations. Each adversary's influence and visibility are defined by its observable projection of the system execution trace τ , as introduced in Section 3.

4.2 Formal Adversary Model

We formalise adversaries as entities that interact with or observe the system through a restricted view of its execution trace. An adversary is represented as a tuple $\mathcal{A} = (\text{View}, \text{Cap}, \text{Obj}, \text{Res})$, where: (1) View specifies the events observable to the adversary, formalised as a projection $\tau \downarrow_{\mathcal{A}}$; (2) Cap captures the adversary's operational capabilities, such as reading, writing, modifying, injecting, or inferring information; (3) Obj specifies the privacy-violating objectives pursued by the adversary; (4) Res denotes resource bounds, including computational power, access to background knowledge, and machine learning inference capabilities.

Adversary classes. We consider the following adversary classes:

- $\mathcal{A}_{ext}$: an external adversary observing or modifying interactions across trust boundaries;
- $\mathcal{A}_{mal}$: a malicious adversary controlling one or more agents in $\mathcal{A}$;
- $\mathcal{A}_{hbc}$: an honest-but-curious agent that follows prescribed protocols while exploiting unintended information leakage;

- $\mathcal{A}_{op}^w, \mathcal{A}_{op}^m, \mathcal{A}_{op}^s$: platform operators with weak, medium, and strong privileges, respectively, ranging from read-only access to full administrative control;
- $\mathcal{A}_{aud}$: an auditor that misuses compliance or oversight access;
- $\mathcal{A}_{3p}$: a third-party service or integration acting adversarially.

Adversarial view over traces. Each adversary observes a potentially partial, delayed, or sanitised projection of the system execution trace $\tau \downarrow_{\mathcal{A}} \subseteq \tau$. The projection operator restricts the events and attributes visible to the adversary. For example, an agent may observe only its own actions and received messages, whereas an operator may observe logs, snapshots of shared memory, or selected components of KB depending on its privilege level.

4.3 Collusion Model

Adversaries may act independently or collude by sharing observations and derived information. Let $\mathcal{S}$ denote the set of adversary classes defined above. A collusion set is any subset $C \subseteq \mathcal{S}$. The collective adversarial view is defined as the union of individual projections

$$\tau \downarrow_C = \bigcup_{\mathcal{A}_i \in C} \tau \downarrow_{\mathcal{A}_i}.$$

We allow arbitrary collusion among adversaries, including combinations of external adversaries, compromised agents, and high-privilege operators. Adversaries may further combine their observations with background knowledge, external datasets, or learned inference models.

4.4 Trust Boundaries and Attack Surfaces

We consider three *trust zones* that determine adversarial access and opportunities for privacy leakage: (a) **internal (trusted)** components under organisational control, including authorised agents and private infrastructure; (b) **boundary and semi-trusted interfaces** that mediate data ingress and egress with external entities, such as application programming interfaces and integration middleware; and (c) **external (untrusted)** users, third-party services, external data sources, and systems outside organisational control. Crossing a trust boundary may increase the granularity or scope of adversarial observations and expand the available attack surface.

The principal privacy-relevant *attack surfaces* are: (a) **inter-agent communication**, including message interception or manipulation, content inference, prompt injection, and coercion of agents into revealing sensitive context; (b) **shared memory and knowledge-base access**, including unauthorised reads or writes to KB that enable leakage, poisoning, or reconstruction of private data or agent state; (c) **autonomy, delegation, and tool use**, through which autonomous task decomposition, delegation, or invocation of external tools may create covert information flows; (d) **logging, tracing, and observability**, where logs, memory snapshots, introspection outputs, or debugging and audit traces may disclose sensitive information; and (e) **privacy-enforcement evasion**, including attempts to bypass, corrupt, or manipulate privacy controls, suppress events, alter logs, or evade detection by Priv.

4.5 Attacker Capabilities and Objectives

Capabilities. We assume that adversaries may possess the following capabilities: (a) inference and attribute extraction from partial observations, outputs, or explanations; (b) reconstruction of personal or sensitive data from responses, memory representations, embeddings, or intermediate artefacts; (c) cross-agent correlation of behaviours and outputs to re-identify individuals or infer latent attributes; (d) compromise or impersonation of agents to extract private information or influence system behaviour; (e) manipulation of prompts, roles, delegation, or tool invocation to induce leakage or propagate sensitive data; and (f) modification, suppression, or injection of trace and log events to influence privacy enforcement or forensic analysis.

Objectives. Adversarial objectives include: (i) unauthorised access to private data, agent state, or contextual information; (ii) inferred disclosure of sensitive attributes through aggregation or indirect signals; (iii) inducing unbounded or policy-violating retention of personal data; and (iv) propagating sensitive information across agents, trust boundaries, or external services in violation of policy.

4.6 Adversarial Knowledge and Observability

Adversaries may observe different *views* of system execution depending on their role, privileges, and access to observability infrastructure. In practice, these views arise from distinct stages of the tracing and logging pipeline rather than from direct access to the full execution trace.

We distinguish conceptually between the raw execution trace, which records all system events but is not directly observable in deployed systems; the logged trace $\tau_{\log}$, which consists of events emitted to monitoring, debugging, or auditing infrastructure; and sanitised views $\tau_{\log}^{\text{san}}$, which apply filtering, aggregation, redaction, or privacy mechanisms before exposure to specific actors. For each adversary $\mathcal{A}$, the effective observable view is determined by access-control policies, logging configuration, and sanitisation mechanisms. When adversaries collude, their joint observational power corresponds to the combination of their respective views, as defined in Section 4.3.

Privacy breach condition. Let φ denote a protected predicate, such as a sensitive attribute of an individual, and let $\text{Adv}_{\mathcal{A}}(\varphi \mid \tau \downarrow_{\mathcal{A}})$ denote the adversary’s advantage in inferring φ from its observable view. A trace violates privacy if there exists an adversary or colluding set whose advantage exceeds a tolerance threshold ϵ :

$$\exists \mathcal{A} \text{ or } C : \quad \text{Adv}_{\mathcal{A}}(\varphi \mid \tau \downarrow_{\mathcal{A}}) > \epsilon \quad \Rightarrow \quad \text{Priv}(\tau) = \text{false}.$$

4.7 Defender Model

Defenders enforce privacy-preserving mechanisms, policy constraints, and compliance checks on agents and shared state. These include (i) policy-governed privacy mechanisms such as information flow control, purpose limitation, consent enforcement, and retention or deletion rules; (ii) privacy-enhancing technologies including differential privacy, secure aggregation, secure multi-party computation, trusted execution environments, cryptographic attestation, zero-knowledge proofs, and selective homomorphic encryption; and (iii) observability hygiene mechanisms such as privacy-aware logging, scoped auditor access, trace integrity protection, and defences against manipulation of Priv.

Defences are evaluated with respect to adversarial trace projections and collusion assumptions defined above. The threat model remains privacy-centric and does not attempt to enumerate all security threats unless they directly affect privacy guarantees.

5 Multi-agent Privacy Sandbox Methodology

5.1 Purpose and Scope of the Revised Sandbox

We implement a multi-agent privacy sandbox with policy-governed agents, tool-mediated actions, persistent state, logging, and adversarial trace projections to evaluate whether the formal leakage taxonomy arises under controlled but realistic multi-agent conditions. The sandbox is not intended to replace public benchmarks or to reproduce any specific deployed clinical system. Instead, it provides a controlled experimental environment for measuring how privacy-relevant information propagates through agent roles, tool calls, memory updates, messages, final outputs, and audit logs.

The purpose of the experiment is to quantify mitigation effects and residual leakage under explicit assumptions. The experiment does not establish a general privacy guarantee for agentic or multi-agent AI systems. Rather, it

tests whether direct disclosure, semantic cues, tool-mediated exposure, and adversarial inference advantage can be measured in a way that is consistent with the formal trace model introduced earlier in the paper.

5.2 Design Goals

The sandbox is designed around four empirical requirements. First, agents can be executed in multiple privacy modes, including uninstructed agents, privacy-instructed agents, and policy-checked agents. This allows the evaluation to distinguish leakage caused by fully naive behavior from leakage that remains when agents are explicitly instructed to minimize disclosure. Second, the execution trace includes tool-mediated events, including `tool_call`, `tool_result`, `external_send`, `memory_update`, `policy_check`, `message`, and `log` events. These trace events model the fact that privacy leakage in agentic systems often occurs through intermediate channels, not only through final user-facing outputs.

Third, leakage is not measured only by exact matching of planted sensitive tokens. The evaluator additionally measures semantic-hint leakage, compositional inference leakage, tool-mediated leakage, policy violations, leakage surface, and adversarial inference advantage over trace projections. Fourth, the simulator supports both a deterministic backend for large reproducible Monte Carlo experiments and a local LLM backend through LM Studio. The deterministic backend supports controlled ablations and repeatability, while the LLM backend validates that the same measurement framework can be applied when agent roles are instantiated through a real instruction-following language model.

5.3 Scenario Families and Calibration to Prior Work

To calibrate the sandbox against related empirical work, we organise the evaluation into five benchmark-style scenario families inspired by prior privacy evaluations: ConfAIde-style contextual privacy, PrivacyLens-style tool-mediated agent privacy, AgentLeak-style internal-channel leakage, MAGPIE-style multi-agent contextual privacy, and compositional-leakage scenarios. The objective is not to reproduce the original datasets, prompts, or leaderboard protocols of these benchmarks. Instead, we use them as reference points for constructing comparable privacy conditions inside our trace-based model.

The calibration proceeds in four steps. First, we identify the privacy phenomenon emphasised by each line of work, such as contextual norm violation, tool-mediated disclosure, internal message leakage, or inference from partial disclosures. Second, we instantiate a corresponding scenario template in the sandbox using the same underlying clinical workflow, agent roles, and synthetic sensitive predicates. Third, we expose the relevant trace channels to adversarial projections, including final outputs, inter-agent messages, memory states, tool calls, logs, and combined views. Fourth, we evaluate all scenario families under the same architectures, privacy modes, episode counts, and leakage metrics. This creates a controlled comparison in which the privacy channel varies by scenario family while the measurement framework remains fixed.

This design provides a common empirical frame for comparing privacy channels without claiming that the sandbox replaces the original benchmarks. In particular, the comparison is not a direct leaderboard comparison against ConfAIde, PrivacyLens, AgentLeak, MAGPIE, or compositional-leakage evaluations. Rather, it is a calibration showing that the sandbox can represent the main privacy channels studied by those works and measure them using the formal trace-projection threat model introduced in this paper.

Table 2 summarises the mapping. The first column identifies the benchmark-style family. The second column states the privacy phenomenon represented. The third column explains how that phenomenon is instantiated in our sandbox. The fourth column lists the trace projections exposed to adversarial evaluation. The final column reports the metrics used for comparison across all families.

Comparison protocol.

Table 2. Alignment of the sandbox with related privacy evaluation settings. The comparison is not a reproduction of the original datasets or leaderboards; instead, each family is operationalised as a controlled scenario template over the same trace model, agent roles, adversarial projections, and leakage metrics.

Scenario family	Privacy phenomenon represented	Sandbox operationalisation	Adversarial views	trace	Primary metrics
ConfAIde-style [21]	Contextual privacy norms and multi-party information flow	Role-conditioned clinical task in which some disclosures are appropriate for one recipient but inappropriate for another. The policy checker evaluates whether information flows respect the intended role and purpose.	Final output; role-conditioned messages; policy-check events		Policy violation rate; direct leakage; inference advantage
PrivacyLens-style [37]	Privacy decisions in tool-mediated language-agent trajectories	Agents decide whether sensitive attributes should be included in tool arguments, tool results, or external sends. Leakage is measured when privacy-relevant information enters a tool channel unnecessarily.	Tool calls; tool results; external sends; final output		Tool leakage; policy violation rate; inference advantage
AgentLeak-style [9]	Leakage through internal multi-agent channels	Sensitive information may propagate through inter-agent messages, shared memory, logs, or intermediate tool traces even when the final output appears safe.	Messages; shared memory; logs; tool traces; combined projection		Any-leak rate; channel-specific leakage; inference advantage
MAGPIE-style [17]	Multi-agent contextual privacy in high-stakes collaborative tasks	Multiple agents coordinate over a clinical workflow with different roles and access needs. The evaluator measures whether delegated actions or messages expose sensitive information beyond the intended context.	Agent-to-agent messages; delegated actions; external recipients		Policy violation rate; trace-projection leakage; inference advantage
Compositional-leakage [28]	Emergent leakage from individually partial disclosures	No single event needs to reveal the sensitive predicate directly. Separate semantic hints across messages, logs, and tool calls can be combined by an adversary to infer the protected attribute.	Semantic hints across messages; logs; tools; combined projections		Semantic-hint leakage; compositional inference leakage; inference advantage

For each scenario family, we evaluate the same two LLM-backed architectures, `privacy_aware_none` and `privacy_aware`, under the same three privacy modes, `none`, `instructed`, and `policy_checked`. Each experimental cell is run for 100 episodes. The same latent sensitive predicate, adversarial inference procedure, and leakage metrics are used across all families. This protocol allows us to compare how different privacy channels behave under a fixed measurement framework. The baseline for comparison is `privacy_aware_none + none`; the strongest protected configuration is `privacy_aware + policy_checked`.

Relation to existing benchmarks.

The benchmark-style families are used to align our sandbox with privacy phenomena studied in existing benchmarks, not to claim leaderboard superiority. ConfAIde, PrivacyLens, AgentLeak, MAGPIE, and compositional privacy benchmarks provide valuable datasets and evaluation protocols for specific privacy tasks. Our sandbox serves a different purpose: it enables controlled structural comparison under a unified Dec-POMDP and trace-based model. In particular, the same latent sensitive predicate, privacy policies, agent roles, adversarial trace projections, and leakage metrics are held fixed across scenario families. This makes it possible to compare how different privacy channels behave under the same architectures and privacy modes. The resulting evaluation is therefore suited to testing the paper’s formal claims about structural leakage, while the original benchmarks remain the appropriate reference points for dataset-specific or leaderboard-style evaluation.

5.4 Agent Roles and Latent State

Each episode instantiates a simplified clinical workflow with four agent roles: *Planner*, *DataAgent*, *Writer*, and *Audit*. The *Planner* decomposes the task and coordinates the workflow. The *DataAgent* retrieves synthetic patient attributes from the latent state. The *Writer* prepares a task output, such as a discharge summary or external communication. The *Audit* agent records workflow events and monitors policy-relevant actions.

The latent state contains synthetic sensitive attributes, such as diagnosis and address fields. These values are not intended to represent real patient data. They are controlled attributes that make it possible to measure direct disclosure, partial semantic disclosure, and adversarial inference from trace projections. Because the ground-truth sensitive predicate is known, the evaluator can estimate whether an adversary can infer the protected attribute from final outputs, inter-agent messages, tool-mediated events, memory states, logs, or combined projections.

5.5 Workflow and Execution Pipeline

The sandbox evaluates each configuration through a common execution pipeline. Each episode begins with a synthetic clinical task and a known latent sensitive predicate. The agent roles then execute the task through messages, tool calls, memory updates, external sends, and audit logging. This produces a structured execution trace that records events such as `message`, `tool_call`, `tool_result`, `memory_update`, `external_send`, `policy_check`, and `log`.

During execution, the selected privacy mode determines whether agents receive no privacy guidance, natural-language privacy instructions, or explicit policy checks before releasing candidate outputs, memory writes, tool arguments, and external sends. The selected architecture determines whether privacy-aware controls such as redaction, tool-call filtering, scoped logging, and privacy labels are enabled.

After execution, the evaluator constructs adversarial trace projections over final outputs, inter-agent messages, memory states, tool-mediated events, logs, and combined views. Leakage is then measured using direct secret disclosure, semantic-hint leakage, compositional inference leakage, tool-mediated leakage, policy violation rate, any-leak rate, leakage surface, and adversarial inference advantage. This pipeline keeps the task environment fixed while varying scenario family, architecture, and privacy mode.

5.6 Privacy Modes

We evaluate three privacy modes that progressively increase the strength of privacy intervention. The first mode, `none`, provides an unprotected baseline in which agents complete the task without privacy-specific instructions or policy enforcement. This setting captures the leakage surface that arises from the workflow structure itself when no explicit privacy control is applied. The second mode, `instructed`, adds natural-language privacy instructions and data-minimisation guidance to the agents. Agents are asked to avoid unnecessary disclosure of sensitive information, but their candidate outputs, memory updates, tool arguments, and external sends are not blocked by a hard policy gate. This mode isolates the effect of instruction-level privacy guidance and tests whether prompting alone reduces disclosure. The third mode, `policy_checked`, combines privacy instructions with an explicit policy-checking layer. Before release, candidate outputs, memory writes, tool arguments, and external sends are checked against a privacy policy. This mode evaluates whether leakage persists even when agents are both instructed to protect sensitive information and constrained by an enforcement mechanism.

5.7 Architecture Contrast for LLM-backed Validation

The empirical validation compares two architectures that use the common execution pipeline described above but differ in the presence or absence of privacy-aware controls. This design isolates the effect of privacy-aware execution while keeping the workflow, agent roles, scenario families, synthetic sensitive predicates, and leakage metrics fixed. The first architecture, `privacy_aware_none`, disables privacy-specific controls. It serves as the unprotected baseline and captures leakage arising from task execution, tool use, memory, and inter-agent coordination when no privacy-aware enforcement is applied. The second architecture, `privacy_aware`, adds redaction, privacy-aware tool-call filtering, policy checks, scoped logging, and privacy labels on trace events. This architecture represents the strongest privacy-aware configuration evaluated in the LLM-backed experiments.

5.8 Adversarial Trace Projections

Consistent with the threat model in Section 4, the evaluator constructs adversarial views over trace projections rather than assuming access to the full raw trace. We evaluate adversaries that observe final outputs, inter-agent messages, shared memory, logs, tool-mediated events, and combined projections.

For a protected predicate ϕ , such as whether the patient has a sensitive diagnosis, adversarial inference advantage is estimated as

$$\text{Adv}_A(\phi \mid \tau \downarrow A) = \Pr[\hat{\phi}_A = \phi \mid \tau \downarrow A] - \Pr[\hat{\phi}_0 = \phi], \quad (1)$$

where $\hat{\phi}_A$ is the adversary’s inferred predicate from its observable projection and $\hat{\phi}_0$ is a prior-only baseline. We report the maximum inference advantage across evaluated projections, together with channel-specific leakage indicators.

This formulation connects the empirical measurement directly to the trace-projection model. A disclosure is therefore not limited to an exact occurrence of a sensitive string. It can also arise when the adversary observes a combination of messages, logs, tool calls, or semantic hints that increases its ability to infer the protected predicate.

5.9 Leakage Metrics

A system may eliminate exact secret-string disclosure while still revealing sensitive information through tools, memory, semantic cues, or multi-agent composition. For each configuration, we report task success, any-leak rate, direct secret leakage, semantic-hint leakage, compositional inference leakage, tool-mediated leakage, policy violation rate, leakage surface, and adversarial inference advantage.

Direct secret leakage corresponds to explicit disclosure of synthetic sensitive values. Semantic-hint leakage captures partial cues, such as oncology-related terms or other medically suggestive attributes, that may not reveal the sensitive value directly but increase inferential risk. Compositional inference leakage is recorded when multiple individually partial cues allow inference of the protected predicate. Tool-mediated leakage captures sensitive information transmitted through tool arguments, tool results, or external sends. Policy violation rate records cases where an action violates the explicit privacy policy. Leakage surface measures how many distinct agents or channels touched privacy-relevant information during an episode.

5.10 Experimental Platform and LLM Validation Setup

To complement the deterministic backend, we conducted an LLM-backed validation using a local inference environment. The purpose of this validation is to test whether the privacy-control behavior observed in the sandbox persists when the agent roles are instantiated through a real instruction-following language model, rather than only through deterministic rules. This experiment is a model-specific validation of the measurement framework, not a general benchmark across LLM families.

The validation was run using LM Studio’s local inference server with the loaded model identifier qwen/qwen3.6-35b-a3b. The model is Qwen3.6-35B-A3B, a mixture-of-experts model with 35B total parameters and approximately 3B active parameters. The reported local configuration used GGUF format with Q4_K_M quantization. The local server was accessed through LM Studio’s REST interface on port 1234. For reproducibility, decoding was performed with temperature 0 and a maximum generation length of 96 tokens. These settings reduce decoding variability while keeping each agent response short enough to associate observed leakage with task-relevant trace events. The LLM validation was run on a Windows 11 workstation equipped with an AMD Ryzen Threadripper PRO 3975WX processor, 128 GB of system memory, and an NVIDIA RTX A5000 GPU. A complete overview of the experimental platform is provided in the supplementary document.

5.11 Experimental Design

The LLM-backed validation evaluates two architectures, three privacy modes, and five benchmark-style scenario families. Each experimental cell is run for 100 episodes, yielding $2 \times 3 \times 5 \times 100 = 3000$ LLM-backed episodes in total.

Statistical reporting. We report the empirical quantities in this controlled validation as descriptive point estimates. The 100 episodes in each experimental cell form a fixed synthetic calibration suite under controlled scenario templates, architectures, privacy modes, and one local LLM configuration; they are not treated as an IID sample from a defined real-world deployment population. Accordingly, the reported rates and inference-advantage values are used for within-design comparison rather than population-level significance claims, and small numerical differences between cells should not be interpreted as statistically significant deployment-level differences. A reported rate of 0.00 means that no event of that type was observed in the 100 episodes of the corresponding cell; it does not establish that the underlying probability is universally zero.

The primary comparison in the results section is between the unprotected baseline, `privacy_aware_none + none`, and the strongest protected configuration, `privacy_aware + policy_checked`. This comparison isolates the effect of privacy-aware policy checking relative to an otherwise comparable workflow substrate. Additional cells involving instructed mode are used to distinguish natural-language privacy guidance from explicit policy enforcement.

The deterministic backend remains available for larger Monte Carlo runs and architecture ablations. We use the LLM-backed results to validate that the measurement framework applies to real instruction-following model outputs, while using the deterministic backend to support reproducible stress testing and ablation analysis.

Additional implementation details, execution commands, output files, and artifact limitations are provided in the supplementary document.

6 Empirical Calibration Results

6.1 Overview of Results

Table 3 reports the LLM-backed validation across all five benchmark-style privacy families. For each family, we compare the unprotected baseline, `privacy_aware_none + none`, with the strongest protected configuration, `privacy_aware + policy_checked`. The baseline disables privacy controls and privacy instructions, while the protected configuration applies privacy-aware policy checking to outputs, tool arguments, memory writes, and external sends.

The protected configuration preserves task success, with no direct secret disclosure or explicit policy violation observed in the evaluated episodes. However, the protected configuration does not eliminate all privacy-relevant leakage. Residual leakage remains visible through tool-mediated channels and semantic hints. We therefore interpret these results as evidence that policy-checked privacy-aware execution reduces direct disclosure and adversarial inference advantage, but does not provide a complete privacy guarantee.

6.2 LLM-backed Empirical Calibration

The primary outcome is maximum adversarial inference advantage over the evaluated trace projections. The unprotected baseline yields high inference advantage because sensitive attributes are directly or indirectly visible through multiple channels. The baseline maximum advantage of 0.4900 is empirically computed under the common unprotected condition. Its identical value across the five scenario families is a consequence of the controlled comparison design: the latent sensitive-predicate distribution, prior-only reference, adversarial inference procedure, and `privacy_aware_none + none` configuration are held fixed across families. The scenario families vary the privacy channel being operationalised, while the common baseline is deliberately retained as a fixed reference. The protected configuration then produces family-specific values because the residual information available through the corresponding trace projections differs across scenario families.

Table 3. LLM validation across all five benchmark-style privacy families. The unprotected condition, `privacy_aware_none + none`, is a common controlled baseline across families; its maximum advantage of 0.4900 is empirically computed, while the repeated value reflects the deliberately fixed baseline construction. The protected configuration is `privacy_aware + policy_checked`. Inference advantage is measured over adversarial trace projections; lower is better. Zero direct-leak and policy-violation entries denote zero observed events in the corresponding 100-episode cells.

Sandbox scenario family	Baseline max advantage	Protected max advantage	Advantage reduction	Protected any leak	Direct leak	Policy violations
ConfAIde-style	0.4900	0.0054	98.9%	0.09	0.00	0.00
PrivacyLens-style	0.4900	0.0090	98.2%	0.85	0.00	0.00
AgentLeak-style	0.4900	0.0126	97.4%	0.69	0.00	0.00
MAGPIE-style	0.4900	0.0108	97.8%	0.52	0.00	0.00
Compositional-leakage	0.4900	0.0216	95.6%	0.51	0.00	0.00
Mean across families	0.4900	0.0119	97.6%	0.53	0.00	0.00

Across the five scenario families, the protected configuration reduces maximum adversarial inference advantage by 95.6–98.9% relative to the unprotected baseline. No direct secret leakage or policy violation was observed in the protected episodes. At the same time, the protected any-leak rate remains non-zero, showing that the evaluation continues to detect residual structural leakage after direct disclosure is blocked.

6.3 Residual Leakage Channels

The results in Table 3 show a large reduction in adversarial inference advantage while residual privacy-relevant signals remain. To separate direct policy failure from structural exposure, Table 4 decomposes the protected configuration by leakage channel. Direct secret leakage and explicit policy violations are zero in the evaluated protected episodes, whereas any-leak remains non-zero because the channel detector also records semantic hints and tool-mediated exposure.

Tool-leak rate and adversarial inference advantage should not be interpreted as equivalent quantities. The tool-leak indicator is deliberately conservative: it records an episode when privacy-relevant information enters a tool argument, tool result, or external-send event, even if that information is partial, generalised, redacted, or insufficient by itself to identify the protected predicate. Inference advantage instead measures whether the evaluated adversary can infer the protected predicate from an observable projection more accurately than from the prior-only baseline. Tool-event projections and the combined trace projection are included in this adversarial evaluation; therefore, the low protected inference advantage is not caused by excluding the tool channel. Rather, many detected tool exposures are weakly discriminative under the evaluated adversary. We nevertheless treat the inference procedure as bounded: a stronger adversary may exploit residual tool-channel cues more effectively. The results therefore support a narrower conclusion—policy checking substantially reduces successful predicate inference and direct disclosure, but it does not eliminate privacy-relevant exposure at intermediate channels.

6.4 Inference Advantage Across Benchmark-style Families

Figure 2 reports maximum adversarial inference advantage for the baseline and protected configurations across the five benchmark-style families. The baseline configuration has high inference advantage because sensitive attributes are directly or indirectly visible in multiple trace projections. The protected configuration substantially reduces this inferential signal in every family, with relative reductions of 95.6–98.9%.

Table 4. Residual leakage channels in the protected configuration `privacy_aware + policy_checked`. Tool leak is a conservative channel-exposure indicator and is not equivalent to successful inference of the protected predicate. Tool-only and combined trace projections are included in the adversarial inference evaluation. Direct secret leakage and explicit policy violations are zero observed events in the evaluated 100-episode cells, while residual tool-mediated and semantic-hint exposures remain observable.

Sandbox scenario family	Any leak	Tool leak	Semantic hint	Compositional inference	Direct leak	Policy violations
ConfAide-style	0.09	0.06	0.03	0.00	0.00	0.00
PrivacyLens-style	0.85	0.85	0.05	0.00	0.00	0.00
AgentLeak-style	0.69	0.66	0.07	0.00	0.00	0.00
MAGPIE-style	0.52	0.48	0.06	0.00	0.00	0.00
Compositional-leakage	0.51	0.46	0.12	0.00	0.00	0.00

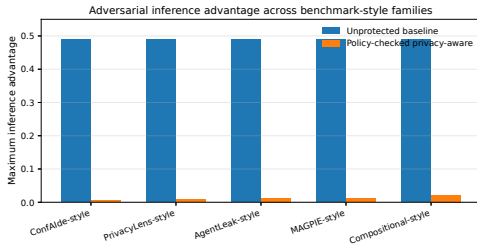


Fig. 2. Maximum adversarial inference advantage across all benchmark-style privacy families. Lower values indicate less inferential signal available to trace-projection adversaries.

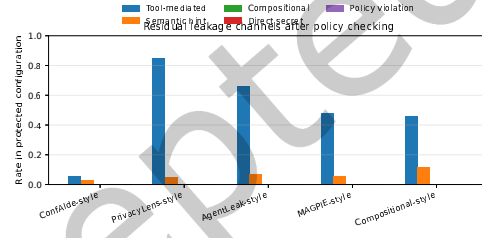


Fig. 3. Residual leakage channels in the policy-checked privacy-aware configuration. No direct secret disclosure was observed in these controlled episodes, while residual semantic and tool-mediated leakage remains.

6.5 Residual Structural Leakage

Figure 3 visualises the residual leakage channels under the protected configuration. Tool-mediated and semantic-hint channels remain visible, while direct secret leakage and explicit policy violations are zero. This pattern supports the paper’s central argument that direct disclosure controls are necessary but insufficient for structural privacy protection in agentic and multi-agent systems.

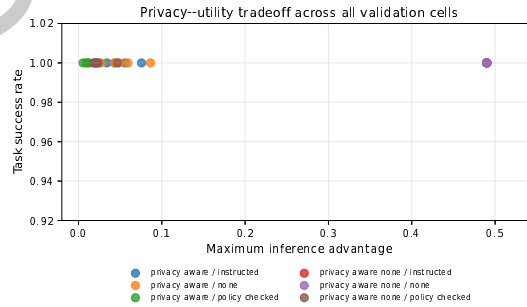


Fig. 4. Privacy-utility tradeoff across all LLM validation cells. Policy-checked privacy-aware agents preserve task success while reducing adversarial inference advantage.

6.6 Privacy–Utility Tradeoff

Task success remains at 1.0 in both baseline and protected configurations in the evaluated scenarios. Figure 4 therefore shows that the observed privacy gains are not caused by agents refusing or failing the task. Instead, the improvement arises from restricting unnecessary disclosure while preserving workflow completion.

6.7 Implications for the PET-to-risk Mapping

The empirical results support the qualitative PET-to-risk mapping in Section 9. Redaction and policy checking reduce direct output leakage. Scoped logging reduces audit exposure. Tool-call filtering reduces, but does not eliminate, tool-mediated exposure. The remaining semantic-hint and tool-mediated leakage illustrates why existing PETs provide only partial coverage for emergent and compositional leakage.

These results should therefore be read as empirical calibration of the structural mismatch identified in Table 5, not as a complete privacy guarantee. They show that direct disclosure controls are helpful but insufficient when information can be reconstructed through long-horizon traces, intermediate tool calls, partial semantic cues, or multi-agent composition.

6.8 Interpretation and Limitations

The principal limitation of the LLM-backed validation is that it uses one local instruction-following model and one quantisation configuration: Qwen3.6-35B-A3B with Q4_K_M quantisation under deterministic decoding. The experiment should therefore be interpreted as model-specific validation that the proposed measurement framework can be instantiated with a real LLM, not as evidence that the measured mitigation effects generalise across model families, parameter scales, quantisations, prompting strategies, or decoding configurations.

Within that scope, the LLM-backed validation supports three bounded conclusions. First, no direct secret disclosure or explicit policy violation was observed under the strongest protected configuration in the evaluated episodes. Second, adversarial inference advantage is substantially reduced relative to the common unprotected baseline across all five benchmark-style families. Third, residual structural exposure remains, particularly through tool-mediated and semantic-hint channels. The first conclusion is intentionally stated as “zero observed events” rather than as a universal zero-leakage guarantee.

The reported empirical quantities are descriptive point estimates for the controlled 100-episode cells. Because the synthetic scenario suite is not treated as an IID sample from a defined deployment population, we do not use these values for population-level significance claims. The sandbox also uses synthetic sensitive attributes and stylised workflows, and the evaluated inference procedure is bounded; a stronger adversary may extract more information from residual semantic or tool-mediated cues. Finally, the benchmark-style families align the sandbox with privacy phenomena studied in related work but do not reproduce the original datasets or scoring protocols. Future work should extend the evaluation to additional models and quantisations, larger and more diverse task suites, stronger inference adversaries, richer operational workflows, human privacy judgments, and direct execution on public benchmark datasets where licensing permits.

7 Taxonomy of Privacy Risks in Agentic and Multi-Agent AI

This section presents a taxonomy of privacy risks that arise specifically in agentic and multi-agent AI systems. Existing privacy taxonomies primarily focus on static machine learning pipelines, federated learning, or inference attacks against isolated models. In contrast, agentic and multi-agent environments introduce distinct privacy risks driven by autonomous decision making, inter-agent coordination, persistent memory, emergent system behaviour, and dynamic task delegation. The taxonomy introduced here captures these properties and organises

privacy risks into five categories, each corresponding to a recurring information leakage pathway observed in agentic and multi-agent executions.

The taxonomy is grounded in the formal system and threat models introduced in Sections 3 and 4. Each category corresponds to characteristic patterns in execution traces τ , adversarial trace projections $\tau \downarrow_X$, and the evolution of agent knowledge states κ_i^t within the hybrid Dec-POMDP framework. By analysing how information propagates through observations, actions, memory updates, shared knowledge bases, and inter-agent communication, we identify systematic classes of privacy leakage that arise from autonomy, coordination, persistence, and interaction over time.

7.1 Taxonomy Overview

Figure 5 summarises the taxonomy, which consists of five main categories: autonomy induced overexposure, coordination and information propagation, memory and persistent state leakage, emergent behaviour and compositional leakage, and observability and audit leakage. These categories are not mutually exclusive. A single privacy violation may involve multiple leakage pathways operating simultaneously or sequentially.

7.2 Autonomy-Induced Overexposure

Agentic systems autonomously plan and execute actions, including accessing data sources and invoking external tools. Autonomy increases privacy risk by enabling behaviours such as: (a) *over-permissioned access*, in which agents access more data than is strictly required for task completion; (b) *unconstrained tool invocation*, where agents disclose sensitive information to external services or application programming interfaces; and (c) *delegation-based exposure*, in which agents delegate tasks without propagating access constraints, consent conditions, or purpose limitations.

From the perspective of the formal model, autonomy induced overexposure arises when an agent policy π_i selects data access or tool invocation actions that expand the agent observable trace projection $\tau \downarrow_{A_i}$ beyond what is necessary for the task. Because these actions are selected autonomously based on the agent local knowledge state κ_i^t , traditional access control mechanisms tied to static workflows or predefined roles are insufficient.

In healthcare settings, this risk arises when a clinical decision agent autonomously queries external diagnostic or summarisation services during planning, without propagating patient consent or purpose constraints. Although each access may appear locally justified, autonomous goal decomposition can lead to disclosure of sensitive clinical attributes outside the original care context.

7.3 Coordination and Information Propagation

Multi-agent coordination enables agents to share intermediate results, context, and decisions. While essential for performance, coordination creates privacy risks including: (a) *inter-agent leakage*, where sensitive data is transmitted between agents without adequate minimisation; (b) *cross-context inference*, where partial observations from multiple agents can be combined to infer protected attributes; and (c) *policy boundary violations*, where coordination crosses organisational or regulatory boundaries without appropriate controls.

Within the hybrid Dec-POMDP model, coordination driven leakage arises when information revealed through inter-agent communication or shared actions accumulates across multiple trace projections. Even if each agent observes only a partial projection $\tau \downarrow_{A_i}$, the aggregation of observations across agents, especially under collusion, can enable inference of sensitive attributes that are not visible in any single interaction.

In healthcare workflows, coordination among agents responsible for diagnosis, resource management, insurance liaison, and patient communication can result in cross-context inference of sensitive conditions, even when each agent reveals only limited information consistent with its operational role.

7.4 Memory and Persistent State Leakage

Agentic and multi-agent systems commonly rely on internal memory and shared knowledge bases to support continuity, efficiency, and long-horizon reasoning. Persistent state introduces privacy risks such as: (a) retention of personal data beyond authorised purposes or legal limits; (b) contamination of shared knowledge bases with sensitive information that becomes visible to additional agents; (c) irreversible logging of sensitive data in audit trails or internal state snapshots.

In the formal model, persistent state leakage is captured by the evolution of agent knowledge states κ_i^t and shared knowledge bases KB^t over time. Because traces record read and write interactions with memory and logs, sensitive information can remain accessible across timesteps even after it is no longer required for task execution.

In healthcare environments, this risk arises when patient specific summaries or eligibility assessments are stored in long-lived memory or shared repositories, enabling secondary access by agents whose roles do not justify exposure to the underlying sensitive data.

7.5 Emergent Behaviour and Compositional Leakage

Interacting agents may exhibit emergent behaviours that are not anticipated at design time. These behaviours can create privacy risks that do not arise at the level of individual agents: (a) unintended side effects arising from feedback loops or implicit coordination; (b) privacy erosion through composition, where individually safe behaviours become unsafe when combined; and (c) collective inference, where agents together infer sensitive information that no single agent could infer alone.

Emergent and compositional leakage arises when privacy violations cannot be attributed to a single agent action or trace segment, but only to the composition of multiple interactions over time. Such leakage is observable only at the level of complete execution traces τ , highlighting the limitations of single agent or stepwise privacy analysis.

In healthcare multi-agent systems, repeated coordination between clinical, operational, and administrative agents can enable collective inference of protected patient characteristics, even when no agent is explicitly designed to infer such attributes.

7.6 Observability and Audit Leakage

Observability mechanisms are necessary for accountability, debugging, and compliance, but they also introduce privacy risks: (a) leakage through logs and traces that record queries, intermediate outputs, or reasoning artefacts; (b) exposure of sensitive information through monitoring metrics and telemetry; and (c) forensic overexposure during post hoc investigation of audit data.

From a trace based perspective, observability and audit leakage occurs when adversaries gain access to enriched projections such as $\tau_{\log}$ or sanitised variants $\tau_{\log}^{\text{san}}$. Although these projections are intended for oversight, they may expose intermediate actions, memory contents, or inferred context that exceed the original disclosure surface.

In regulated healthcare environments, extensive audit trails are required for compliance. However, detailed traces capturing agent actions and intermediate artefacts can inadvertently expose sensitive patient data to auditors or operators whose access extends beyond the original clinical need.

7.7 Privacy Risks Across the Taxonomy

We now illustrate how each of the five privacy risk categories introduced in Section 7 manifests in the HospitalNet deployment.

Autonomy-Induced Overexposure: The clinical decision agent (CDA) is granted broad read access across medical specialties and care sites in order to avoid missed interactions, such as adverse drug interactions between treatments prescribed by different departments. Over time, this design choice leads to scope expansion, whereby

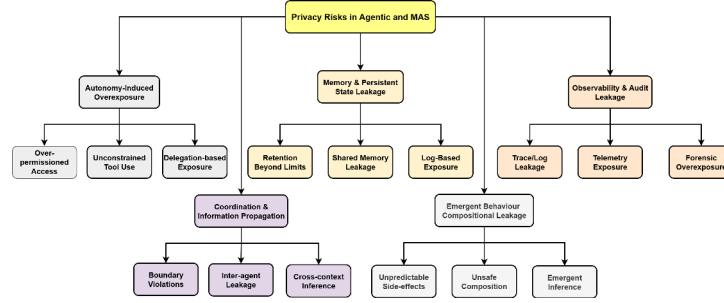


Fig. 5. Taxonomy of privacy risks specific to agentic and multi-agent AI systems.

the agent accesses highly sensitive portions of the medical record that are not strictly necessary for the immediate decision task. Examples include historical reproductive health records or genetic test results that marginally improve predictive accuracy but exceed the original disclosure intent.

Similarly, the patient engagement agent (PEA) may be permitted to query parts of the medical record in order to generate personalised explanations or follow-up prompts. This autonomy to retrieve and combine information increases the likelihood that sensitive data is exposed in contexts where clinicians would ordinarily exercise more restrictive judgement.

Coordination and Information Propagation: The resource coordination agent (RCA) observes aggregated but fine-grained operational signals, including ward occupancy trends, urgency of bed requests, and deployment of specialised equipment. When CDA escalates a case for rapid evaluation, RCA may respond by reallocating monitored beds, assigning staff with specific expertise, or prioritising specialised devices.

Even when RCA is not authorised to access detailed diagnosis codes, changes in resource allocation patterns can leak information. For example, relocation to an isolation ward, mobilisation of specialised equipment, or assignment of specialist staff may signal the presence of sensitive conditions. These signals can be observed by other agents and, in some organisations, by billing or logistics systems, enabling cross-context inference across organisational roles.

Memory and Persistent State Leakage: The shared state store in HospitalNet retains several categories of information, including detailed interactions between clinicians and CDA, PEA chat transcripts and symptom histories, and historical RCA allocation decisions. Although the operational need for some of this data is short-lived, it may be retained indefinitely for purposes such as model improvement or audit. This creates a long-term record from which sensitive conditions, patient behaviours, and clinician practice patterns can be reconstructed.

In multi-tenant deployments, such as cloud-hosted platforms shared across multiple hospitals, additional risks arise. Embeddings, cached representations, or fine-tuned model parameters may inadvertently encode patient-level information that can be probed or inferred by other tenants.

Emergent and Compositional Leakage: Individually, CDA, RCA, and PEA may each be configured with access controls that appear reasonable in isolation. However, when composed through orchestration, their interactions can give rise to new leakage channels. For example, PEA encourages patients to describe symptoms in free text, CDA incorporates this information into risk assessments, and RCA adjusts staffing or bed allocation in response. A staff member with access only to a high-level resource dashboard may infer that certain patients are on end-of-life pathways based solely on repeated allocation patterns, without access to underlying clinical data.

Such emergent leakage arises from the composition of agent behaviours over time rather than from any single action. These dynamics are not captured by traditional component-level threat models but are naturally exposed by a trace-based analysis of multi-agent interactions.

Observability and Audit Leakage. HospitalNet enables extensive logging to satisfy regulatory and safety requirements. This includes CDA recommendations, clinician overrides, PEA escalations, and RCA allocation decisions, all recorded with timestamps and identifiers. These logs are accessible to internal auditors, quality improvement teams, and in some cases external regulators or vendors.

Although intended to support accountability, these logs constitute rich observational traces. An auditor who observes only CDA recommendation summaries and RCA allocation events may still infer sensitive diagnoses or prognoses by correlating the timing, location, and sequence of actions. In terms of the formal model, the adversarial projection $\tau \downarrow \mathcal{A}_{aud}$ may reveal substantially more information than suggested by raw access permissions alone.

7.8 PET Deployment and Residual Gaps

HospitalNet deploys several privacy-enhancing technologies to mitigate the identified risks: (a) *federated learning* with secure aggregation for selected CDA and PEA models, so that raw patient data remains on premises and only encrypted model updates are shared; (b) *differential privacy* for selected analyses of historical data, including incidence statistics and model-evaluation dashboards; (c) *trusted execution environments* for sensitive cloud-based computation, including cross-site training and orchestration logic; (d) *application-layer information-flow control* policies that restrict the data fields each agent may read or write and bind access to authorised purposes such as clinical care, billing, or research; and (e) *privacy-aware logging* that redacts identifiers and sensitive values before log ingestion and applies access controls to log visibility.

Consistent with the mapping in Table 5, this case study highlights several limitations. The deployed PETs are effective at reducing some forms of autonomy-induced overexposure and memory leakage, such as limiting accumulation of raw data in centralised training pipelines. They provide partial mitigation for coordination and propagation risks, for example by preventing direct access to per-patient model updates. However, they offer limited protection against emergent and compositional leakage, as well as against inference from rich observability data. These gaps correspond to the uncovered entries for emergent risks in Table 5.

7.9 Illustrative Leakage as a Trace Violation

We illustrate a concrete leakage pathway using the formal framework. Consider a patient p monitored at home through PEA. Over several days, PEA records repeated episodes of breathlessness and chest pain and escalates these observations to the care team. CDA updates the patient risk score and recommends urgent admission. RCA then performs the following three actions: (a) reserves a high-dependency bed; (b) assigns a nurse with cardiology expertise; (c) relocates specialised monitoring equipment to the relevant ward.

Let τ denote the trace segment containing PEA escalation events, a CDA urgent admission recommendation, the corresponding RCA allocation actions, and associated log entries. An internal auditor X lacks access to diagnosis codes and raw PEA transcripts, but observes a projection $\tau \downarrow_X$ consisting of timestamped CDA recommendations and RCA allocation metadata. From repeated occurrences of this pattern, X can infer with high confidence that patient p is likely experiencing a serious cardiac condition.

The intended policy $\text{Priv}(\tau)$ specifies that adversaries without access to clinical content should not be able to infer sensitive diagnoses. For this trace, however, $\text{Priv}(\tau)$ is violated. The breach arises not from direct disclosure, but from the composition and timing of agent actions observable in the trace. Notably, none of the deployed privacy-enhancing technologies are designed to conceal this form of behavioural signal.

Addressing this gap requires privacy-enhancing mechanisms that operate at the level of coordinated agent behaviour and trace-level observability rather than solely on static datasets or individual services.

8 Privacy-Enhancing Technologies Relevant to Agentic and Multi-Agent AI

This section analyses privacy-enhancing technologies that are relevant to mitigating the risks identified in Section 7. Many existing PETs were developed for traditional machine learning pipelines, distributed analytics, or secure communication settings. Their effectiveness in agentic and multi-agent systems varies significantly due to differences in autonomy, coordination, persistent memory, and emergent behaviour. We group PETs into five classes: input and output privacy, distributed and collaborative privacy, computation privacy, policy-governed privacy, and accountability oriented privacy. For each class, we describe the core mechanisms and analyse their suitability and limitations in agentic and multi-agent environments.

8.1 Input and Output Privacy Mechanisms

Input and output privacy mechanisms aim to limit disclosure through agent inputs, outputs, and externally visible artefacts.

8.1.1 Differential Privacy. Differential privacy provides formal guarantees that the output of a computation does not significantly depend on any single individual's data. Noise can be added to agent outputs, summaries, or logs, with local variants applied at the agent level and central variants applied to aggregated results. Differential privacy has been widely studied for statistical databases, machine learning training, and privacy-preserving analytics, including applications to logging and telemetry.

In agentic and multi-agent systems, differential privacy offers partial protection against output based and audit related leakage. However, its effectiveness is constrained by structural properties of agentic systems. Autonomous agents generate correlated outputs across time and across agents, which violates standard independence assumptions and leads to rapid consumption of privacy budgets. Differential privacy also does not prevent sensitive information from being stored in agent memory or propagated through coordination. As a result, it is best suited for sanitising externally exposed outputs and observability data rather than governing internal agent reasoning or persistent state.

8.1.2 Private Information Retrieval and Query Privacy. Private information retrieval and related query privacy techniques allow agents to access data without revealing query contents or intent. Query obfuscation techniques can reduce leakage of sensitive context associated with autonomous data access. These approaches mitigate some forms of autonomy-induced overexposure but do not address downstream propagation once information has been obtained by an agent.

8.2 Distributed and Collaborative Privacy Mechanisms

Distributed and collaborative privacy mechanisms support privacy-preserving interaction among multiple agents or organisations.

8.2.1 Federated Learning with Secure Aggregation. Federated learning enables model training across distributed data sources without centralising raw data. Secure aggregation ensures that only aggregated model updates are visible to the coordinating entity. Variants such as personalised or hierarchical federated learning can support heterogeneous agent populations.

In agentic systems, federated learning can reduce direct data centralisation and mitigate some memory related risks. However, autonomous agents may bypass federated workflows by directly sharing data, storing sensitive information in memory, or coordinating outside the training pipeline. Federated learning therefore addresses a narrow subset of risks and does not govern agent interaction, delegation, or emergent behaviour.

8.2.2 Secure Multi-Party Computation. Secure multi-party computation enables multiple agents to jointly compute a function over private inputs without revealing those inputs to one another. It provides strong guarantees

against direct inter-agent data leakage during structured joint computation and aligns well with coordination related risks.

However, secure multi-party computation typically assumes fixed participants, predefined functions, and synchronised execution. Agentic systems frequently involve dynamic delegation, adaptive task decomposition, and unstructured reasoning that cannot be expressed as static circuits. As a result, secure multi-party computation is best suited for narrowly scoped collaborative subroutines rather than general multi-agent coordination.

8.2.3 Secret Sharing and Threshold Cryptography. Secret sharing and threshold cryptographic schemes distribute sensitive values across multiple parties so that no single agent can reconstruct the data alone. These techniques provide resilience against compromised agents and collusion but introduce coordination overhead and latency. Their applicability is therefore limited in highly dynamic or real time agentic workflows.

8.3 Computation Privacy Mechanisms

Computation privacy mechanisms protect data during execution.

8.3.1 Trusted Execution Environments. Trusted execution environments provide hardware enforced isolation for sensitive computation and support remote attestation of code and configuration. They have been applied to secure inference, privacy sensitive analytics, and policy enforcement components.

In agentic systems, trusted execution environments can protect specific modules such as memory stores or policy enforcement logic. However, they do not address leakage arising from inter-agent communication, delegation, or emergent behaviour across execution boundaries. They also provide limited transparency into how protected computation interacts with external agents and tools, which limits their usefulness as a comprehensive privacy governance mechanism.

8.3.2 Homomorphic Encryption. Homomorphic encryption enables computation over encrypted data, with fully homomorphic schemes supporting arbitrary computation. While homomorphic encryption offers strong confidentiality guarantees, its computational cost and latency make it difficult to deploy in real time agentic systems with frequent interaction and coordination.

8.4 Policy-Governed Privacy and Information Flow Control

Policy governed privacy mechanisms enforce constraints on data access and propagation.

8.4.1 Information Flow Control. Information flow control enforces confidentiality and integrity constraints on how information propagates between components. Label based and decentralised models are well suited to multi principal settings and shared memory architectures.

In agentic systems, information flow control aligns closely with risks arising from coordination and delegation, as it provides explicit control over data propagation between agents, memory, and trust boundaries. However, applying information flow control to autonomous agents remains challenging due to evolving goals, dynamic policies, and the difficulty of labeling inferred or emergent knowledge. Significant adaptation is required to support compositional privacy guarantees in multi-agent environments.

8.4.2 Zero-Knowledge Proofs. Zero-knowledge proofs enable agents to demonstrate compliance with privacy or usage policies without revealing underlying data. They are particularly useful for audit and accountability, allowing agents to prove that actions or accesses satisfy declared constraints. Zero-knowledge proofs do not prevent leakage during autonomous reasoning or coordination and therefore do not address several core risks identified in Section 7. Their primary value lies in post hoc verification and compliance rather than real time leakage prevention.

8.4.3 Purpose Binding and Consent Enforcement. Purpose binding mechanisms associate data with explicit usage constraints and enforce consent conditions through sticky policies, cryptographic enforcement, or runtime monitoring. These approaches help limit misuse of sensitive data but face challenges in agentic systems where tasks are decomposed and delegated dynamically. Ensuring that consent and purpose constraints propagate correctly under delegation remains an open problem.

8.5 Accountability and Auditable Privacy Mechanisms

Accountability mechanisms aim to support oversight while limiting secondary disclosure.

8.5.1 Privacy-Preserving Logging and Auditing. Privacy preserving logging techniques use redaction, aggregation, encryption, or noise addition to limit the exposure of sensitive information in logs. Verifiable logging structures can support accountability without revealing raw data. In agentic systems, logging remains a major source of leakage because traces capture intermediate actions, coordination patterns, and inferred context. Even sanitised logs can enable inference through correlation over time, which limits the effectiveness of existing approaches.

8.5.2 Transparency and Explanation Mechanisms. Transparency and explanation mechanisms aim to make agent decisions understandable to users and auditors. Many explanation techniques risk exposing sensitive information or internal state. Privacy preserving explainability remains an emerging area and is particularly challenging in multi-agent settings where explanations may reflect collective behaviour rather than individual decisions.

Overall, existing privacy-enhancing technologies provide valuable building blocks but do not fully address the challenges introduced by agentic autonomy, persistent memory, coordination, and emergent behaviour. In Section 9, we systematically map these technologies to the privacy risks identified in Section 7 and highlight gaps that motivate the need for new, trace-aware and interaction-level privacy mechanisms.

9 Analysis and Mapping of PETs to Risks

This section evaluates how the privacy-enhancing technologies (PETs) introduced in Section 8 mitigate the privacy risks defined in Section 7. The goal is not to catalogue techniques, but to analyse coverage and limitations in agentic and multi-agent settings where autonomy, coordination, persistent state, and interaction over time create leakage pathways that differ from those in static machine learning pipelines.

The mapping is grounded in the formal system and threat models in Sections 3 and 4. We assess each PET by the system components it constrains, including agent observations, actions, memory updates, shared knowledge bases, and the adversarial trace projections $\tau \downarrow_X$ that determine what can be observed and inferred. PETs that assume isolated computations or fixed collaboration structures provide guarantees only over restricted fragments of the execution trace. In contrast, persistent-state leakage and emergent compositional leakage arise from the evolution and composition of traces over time and across interacting agents, which many existing technologies were not designed to govern.

9.1 Rationale for PET-to-risk Ratings

Table 5 uses three qualitative ratings and provides a cell-level rationale. A strong rating ($\checkmark$) indicates that the PET directly constrains the trace channel responsible for the corresponding leakage class. A partial rating (Δ) indicates that the PET protects a bounded component, computation phase, or release interface, but does not control downstream propagation through memory, tools, logs, or inferred attributes. A weak rating ($\times$) indicates that the PET does not directly constrain the relevant adversarial trace projection.

The ratings are therefore assigned according to the relationship between each PET and the trace model introduced in Section 3. For a leakage class L , a PET receives strong coverage only when it limits the adversary's observable projection $\tau \downarrow_A$ over the events that instantiate L . For example, a mechanism that encrypts data

during computation may protect data access events, but it does not necessarily constrain later message events, memory writes, tool calls, or audit logs derived from the same sensitive attribute. This distinction explains why several PETs receive partial rather than strong ratings in agentic and multi-agent settings.

9.2 Privacy-budget Composition under Repeated and Correlated Agent Outputs

Differential privacy provides formal guarantees for specific randomized releases, but agentic and multi-agent systems complicate the interpretation of those guarantees because releases may be repeated, adaptive, and correlated across agents. Under basic sequential composition, if a sequence of k mechanisms $M_1, \dots, M_k$ each satisfies ϵ_i -differential privacy for the same protected record, then the composed interaction satisfies at most $\sum_{i=1}^k \epsilon_i$ -differential privacy [7, 8]. Thus, even if each individual release is privacy-preserving, repeated interaction can consume the privacy budget over time.

In an agentic system, the number of releases is not necessarily fixed in advance. Agents may query tools adaptively, write intermediate summaries to memory, retrieve those summaries in later tasks, and disclose related information through logs or inter-agent messages. If an agent releases k outputs with a uniform budget ϵ per release, the basic composition bound becomes $k\epsilon$. For example, ten releases each using $\epsilon = 0.1$ yield a cumulative basic-composition bound of $\epsilon = 1.0$. In a long-horizon workflow, this cumulative loss can grow as agents repeatedly access or summarize the same sensitive predicate.

The multi-agent case is more challenging because different agents may release correlated views of the same latent sensitive attribute. Suppose agents $A_1, \dots, A_n$ each release a sanitized output about the same patient attribute. Even if each release is individually protected, an adversary observing a combined projection $\tau \downarrow X$ may aggregate information across agents, tools, logs, and memory states. In the worst case, privacy loss accumulates across both time and agents:

$$\epsilon_{\text{total}} \leq \sum_{t=1}^T \sum_{i=1}^n \epsilon_{i,t},$$

where $\epsilon_{i,t}$ is the privacy budget consumed by agent i at interaction step t . This expression is a conservative upper bound, but it illustrates why independent per-agent sanitization is insufficient when agents repeatedly disclose correlated information about the same protected predicate.

Correlation also affects inference even when outputs do not contain the exact sensitive value. For example, one agent may release a noisy diagnosis category, another may release a treatment recommendation, and a third may log a specialist referral. Each event may appear individually low-risk, but their combination can increase adversarial confidence about the protected attribute. This is precisely the type of residual leakage captured by the semantic-hint and compositional-inference metrics in Section 6.

This analysis explains why differential privacy receives only partial coverage in Table 5. DP is appropriate for bounded releases such as aggregate telemetry, sanitized audit statistics, or controlled analytics. However, DP does not by itself enforce purpose limitation, prevent raw memory retention, control tool-mediated propagation, or block inference from repeated correlated outputs. In agentic systems, DP must therefore be combined with trace-level budget accounting, memory-retention controls, policy checks over tool use and delegation, and adversarial auditing over trace projections.

9.3 Assessment of Coverage

The mapping yields five observations that explain why coverage is uneven across the taxonomy.

Limited protection against emergent leakage. Emergent and compositional leakage remains largely unmitigated. Most PETs operate on bounded computations, single components, or predefined interaction patterns. They do not constrain privacy loss that arises only from repeated interaction, feedback loops, and multi-step

Table 5. PETs to privacy risks ratings in agentic and multi-agent systems and their rationale.

PET Category or Mechanism	Autonomy Overexposure	Coordination Leakage	Memory Leakage	Emergent Leakage	Audit Leakage
Differential Privacy (Local or Central) [43]	Δ: Bounds leakage from selected randomized releases, but does not stop an autonomous agent from accessing or requesting unnecessary raw data.	Δ: Can sanitize aggregate or released coordination signals, but repeated inter-agent disclosures may compose and reveal correlated information.	×: Does not by itself prevent raw sensitive values or derived summaries from being stored in persistent memory.	×: Does not prevent semantic reconstruction from multiple correlated outputs unless all releases are jointly budgeted and modeled.	✓: Provides strong protection when audit statistics or telemetry are released through a formal DP mechanism.
Private Information Retrieval and Query Privacy	✓: Hides which record or item an agent queries, reducing exposure caused by autonomous retrieval.	×: Does not control what queried information is later shared between agents.	×: Does not govern storage or retention of retrieved content after the query completes.	×: Does not prevent inference from later outputs, messages, or partial semantic cues.	×: Does not sanitize logs or audit traces unless combined with privacy-preserving logging.
Federated Learning with Secure Aggregation [10]	Δ: Reduces exposure of local training updates, but does not restrict autonomous runtime access or output disclosure.	✓: Strongly protects individual updates during aggregation, which directly mitigates coordination leakage in collaborative training.	×: Does not protect agent memory, prompts, retrieved records, or stored intermediate summaries.	×: Does not prevent inference from model outputs, agent messages, or combined post-training behavior.	×: Does not directly protect audit logs or execution traces.
Secure Multi-Party Computation [42]	Δ: Protects inputs during a specified computation, but does not prevent agents from requesting excessive data before or after it.	✓: Strongly protects bounded joint computations by hiding private inputs from coordinating parties.	×: Does not control persistent storage of inputs, outputs, or derived summaries outside the computation.	×: Does not prevent agents from combining outputs with other trace events to infer sensitive predicates.	×: Does not sanitize logs unless the logging mechanism itself is included in the secure computation.
Secret Sharing and Threshold Cryptography [15]	×: Protects secret reconstruction thresholds, but does not control autonomous over-collection or over-sharing.	✓: Strongly mitigates coordination leakage when no single party can reconstruct the protected value alone.	×: Does not prevent agents from storing shares, reconstructions, or derived summaries insecurely.	×: Does not address semantic inference from partial non-secret disclosures.	×: Does not protect audit logs unless logs are separately secret-shared or sanitized.
Trusted Execution Environments [38]	Δ: Can isolate sensitive computation, but authorized code may still over-request or over-disclose outputs.	Δ: Can protect inter-agent computation inside an enclave, but not messages or outputs after enclave release.	×: Does not solve long-term memory leakage if sensitive results are written outside the trusted boundary.	×: Does not prevent inference from permitted outputs, hints, or cross-agent behavior.	Δ: Can protect log generation or attestation inside the enclave, but raw logs may still leak if exposed.
Homomorphic Encryption [22]	×: Protects encrypted computation, but does not constrain autonomous access decisions or output disclosure.	Δ: Supports privacy-preserving computation across parties, but only for the encrypted computation boundary.	×: Does not govern decrypted outputs, persistent memory writes, or retained intermediate summaries.	×: Does not prevent semantic or compositional inference from released results.	×: Does not sanitize logs or audit traces outside the encrypted computation.
Information Flow Control [11]	✓: Labels and policies can directly restrict whether autonomous actions may access or transmit sensitive data.	✓: Directly constrains propagation across messages, tools, and agent boundaries when labels are enforced.	Δ: Can label stored data, but practical enforcement over persistent LLM memory and derived summaries remains difficult.	Δ: Can restrict explicit flows, but implicit, semantic, and inferred flows remain only partially covered.	Δ: Can label logs and audit events, but logs may still reveal metadata or derived contextual information.
Zero-Knowledge Proofs [40]	Δ: Can prove authorization or compliance without revealing witnesses, but does not control all autonomous data use.	✓: Strongly supports verifiable coordination when agents need to prove properties without revealing private inputs.	×: Does not control what agents store in memory after a proof is generated or verified.	Δ: Can reduce disclosure for specific claims, but does not prevent inference from other trace events.	✓: Strong for auditability when compliance can be proven without exposing underlying sensitive data.
Purpose Binding and Consent Enforcement [35]	✓: Directly limits autonomous access and disclosure to authorized purposes and consent conditions.	Δ: Can restrict inter-agent sharing by purpose, but enforcement may weaken across delegated or external tools.	Δ: Can impose retention and reuse limits, but requires reliable memory governance and deletion mechanisms.	×: Does not by itself prevent inference from permitted partial disclosures.	×: Does not protect logs unless audit records are also purpose-bound and minimized.
Privacy-Preserving Logging and Auditing [26]	×: Does not prevent agents from over-accessing or over-sharing during task execution.	×: Does not directly prevent inter-agent propagation before logging occurs.	Δ: Can reduce leakage from retained logs and audit memory, but not from all persistent agent memory.	×: Does not prevent semantic reconstruction from messages, tools, or outputs.	✓: Directly targets audit leakage by minimizing, sanitizing, or protecting audit records.

composition across agents. Consequently, existing PETs provide at most partial mitigation for this category, typically through indirect effects such as restricting information flow or reducing log content.

Partial mitigation of coordination leakage. Distributed PETs such as secure aggregation and secure multi-party computation reduce direct disclosure during collaborative computations. However, they do not address privacy loss through behavioural signals, indirect inference, memory contamination, or downstream propagation after a computation completes. Information flow control and zero-knowledge proofs offer stronger conceptual foundations for governing inter-agent flows because they directly constrain what information may propagate or what can be verified without disclosure, but they require integration with agent execution and policy enforcement.

Weak coverage for memory and persistent state leakage. Most PETs do not directly govern sensitive data stored in internal memory, shared knowledge bases, or long-lived logs. Differential privacy can sanitise selected outputs and log views but does not prevent retention of raw sensitive content in persistent state. Purpose binding and information flow control provide partial mitigation by constraining access and propagation, but practical implementations that support evolving goals, delegation, and inferred knowledge remain limited.

Autonomy-induced overexposure requires policy and control at decision time. Private information retrieval and query privacy can reduce leakage associated with query intent and access patterns. However, autonomy-induced overexposure stems from agent policies that decide what to retrieve, where to send it, and how to delegate tasks. Mitigation therefore requires mechanisms that constrain decisions at runtime, such as purpose binding, consent enforcement, and information flow control integrated into agent planning and tool use, rather than relying solely on cryptographic protection of data transfer.

Strongest support for audit leakage is confined to observability channels. Audit leakage has comparatively strong support through privacy-preserving logging, differential privacy for telemetry, and zero-knowledge proofs for compliance. These approaches protect specific observability projections such as $\tau_{\log}$ and sanitised views. However, they do not fully address inference from correlations in event timing and patterns, which can remain informative even after redaction or noise.

9.4 Structural Gaps and Mismatches

The mapping reveals four recurring mismatches between classical PET assumptions and agentic system properties. First, many PETs assume **bounded computation** with fixed participants, data flows, and execution boundaries, whereas autonomous delegation and tool use can expand these boundaries at runtime. Second, privacy guarantees exhibit **compositional fragility** when outputs are correlated across repeated interactions, multiple agents, and colluding parties. Third, PETs often protect raw data but provide limited **governance of derived information**, including summaries, embeddings, behavioural signals, and inferred attributes that are difficult to control using static policies. Finally, mechanisms such as secure multi-party computation, homomorphic encryption, zero-knowledge proofs, and trusted execution environments may introduce **latency and computational overhead** that is incompatible with interactive or real-time autonomous workflows.

These limitations reflect a broader mismatch between PETs designed for stable, well-scoped computations and agentic systems characterised by autonomous decision-making, persistent state, and evolving interactions. Effective protection therefore requires combining PETs with privacy-aware architectures, runtime policy enforcement, and trace-level governance across agents and over time. Section 11 discusses corresponding research directions for PETs that are trace-aware, delegation-aware, and robust under multi-agent composition.

10 Regulatory and Governance Implications

Agentic and multi-agent AI systems complicate regulatory compliance because privacy-relevant disclosures may occur across delegated tool calls, persistent memory, logs, inter-agent messages, external sends, and inferences

over long execution traces. Such distributed information flows challenge assumptions of clear data controllers, stable purposes, bounded processing, and auditable access paths. Table 6 summarises the principal implications.

Table 6. Regulatory and governance implications for agentic and multi-agent systems.

Regulation Implications	
GDPR	Agentic workflows implicate purpose limitation, data minimisation, storage limitation, transparency, and accountability. Data collected for one purpose may be transformed, retained, or reused in delegated tasks, producing trace-level violations even when individual accesses appear justified. Controls should therefore bind observations, memory writes, tool calls, and external sends to authorised purposes and recipients.
HIPAA	Agentic healthcare systems may violate the minimum-necessary principle by retrieving or sharing more protected health information than an agent’s role requires. Logs may also expose patient attributes, tool arguments, or intermediate reasoning. Governance should therefore combine role-scoped access, purpose-aware disclosure checks, and privacy-preserving auditing.
EU AI Act	Agentic systems used in healthcare or other consequential settings may be subject to high-risk requirements for logging, human oversight, risk management, data governance, and post-market monitoring. In multi-agent deployments, these obligations extend to delegated actions, tool use, memory updates, inter-agent messages, and audit traces, requiring execution-level rather than output-only assessment.

11 Open Challenges and Future Research Directions

Current privacy-enhancing technologies provide incomplete protection for agentic and multi-agent AI systems. Although effective in bounded computations and static workflows, they do not fully address autonomous decision-making, dynamic coordination, persistent state, or emergent behaviour. Tables 7 and 8 outline the resulting challenges and research directions for privacy mechanisms that operate across agent behaviour, interaction, and system evolution.

Table 7. Open challenges and future research directions for privacy in agentic and multi-agent AI systems.

Challenge	Future research direction
Autonomous behaviour and delegation. Planning, task decomposition, delegation, and tool use dynamically expand computation boundaries, weakening controls based on fixed scopes and predefined access. Privacy violations may therefore occur despite local policy compliance.	Embed data-minimisation and purpose-limitation constraints into agent planning. Delegation-aware mechanisms should propagate consent, use, and retention requirements across agents and tools, with runtime verification of permissible information sharing.
Privacy-preserving multi-agent coordination. Repeated and context-dependent communication creates correlated disclosures across agents and over time, limiting privacy guarantees designed for static joint computations.	Develop privacy-preserving coordination using information-flow control, secure communication, and verifiable policy enforcement. Privacy accounting should capture cumulative leakage, correlated inference, and shared responsibility across agents.
Shared memory and persistent state. Sensitive information retained in agent memory, vector stores, logs, or shared knowledge bases may outlive its intended purpose or become accessible to unauthorised entities.	Design memory architectures that enforce access, purpose, retention, and deletion constraints. Further work is needed on verifiable lifecycle governance and privacy-preserving summarisation that preserves utility while reducing sensitive content.
Emergent and compositional leakage. Privacy leakage may arise from combinations of agents, observations, and interactions rather than from any single action, limiting single-agent analysis and static enforcement.	Develop compositional methods over multi-agent execution traces that account for correlation, feedback, and inference over time. Runtime monitoring and privacy-aware adaptation should detect and reduce risky interaction patterns.

Table 8. Open challenges and future research directions (continued).

Challenge	Future research direction
Privacy, auditability, and accountability. Logs, traces, and monitoring support accountability but may disclose sensitive information. Providing verifiable oversight without creating additional leakage remains difficult.	Develop privacy-preserving auditing through sanitised logs, integrity-protected records, and cryptographic evidence of compliance. Agent actions should be verifiable through attestations, signatures, and machine-checkable policies, while accountability models should assign responsibility for privacy violations arising from joint behaviour.
Deployment and performance constraints. The computation, latency, and memory costs of many PETs may be incompatible with interactive or real-time agentic workflows.	Develop lightweight mechanisms with bounded costs for frequent actions and tool calls. Hardware acceleration may support stronger cryptographic and secure-execution guarantees, while adaptive mechanisms should select PETs according to privacy risk, utility, and performance constraints.
Cross-domain and cross-organisational privacy. Multi-agent systems may span organisations and jurisdictions with different trust assumptions, policies, and legal requirements that existing PETs do not adequately address.	Design interoperable policy representations for cross-domain exchange, verification, and enforcement. Distributed trust mechanisms should verify agents operated by different principals, while PETs should support coordination without assuming uniform governance or centralised control.

12 Literature Review

Building on the taxonomy in Section 7, this section reviews prior research on privacy and information protection in agentic and multi-agent AI systems. Rather than attempting exhaustive coverage, we organise representative lines of work according to the five leakage categories introduced earlier: autonomy-induced overexposure, coordination and information propagation, memory and persistent state leakage, emergent behaviour and compositional leakage, and observability and audit leakage. For each category, we summarise how existing methods align with the underlying leakage mechanism and where their assumptions limit effectiveness in agentic and multi-agent settings.

12.1 Contextual Privacy and Agent Privacy Benchmarks

Recent work has begun to evaluate privacy behavior in LLMs and agentic workflows through contextual norms, tool use, and multi-party information flow. ConfAId evaluates privacy norm awareness in LLMs using contextual integrity and multi-party information-flow scenarios [21]. PrivacyLens studies contextual-integrity-grounded privacy decisions in tool-mediated language-agent trajectories [37]. AgentLeak evaluates full-stack leakage across multi-agent workflows and emphasizes internal channels such as messages, memory, tools, and logs [9]. MAGPIE focuses on multi-agent contextual privacy evaluation in high-stakes collaborative tasks [17]. Related compositional privacy work shows that privacy leakage can arise from combining individually partial disclosures across agents or interaction rounds [28].

These works are highly relevant to our study, but their goals differ from ours. They primarily provide benchmark tasks, datasets, or evaluation protocols for particular privacy phenomena. Our contribution is complementary: we provide a formal Dec-POMDP and trace-based abstraction that places autonomy, delegation, coordination, persistent memory, tools, logs, and adversarial projections in a single model. The sandbox then uses benchmark-style families to calibrate how these privacy phenomena appear under the same agent roles, sensitive predicate, privacy modes, architectures, and inference-advantage metrics. Thus, the paper does not claim to supersede these benchmarks; it uses them as reference points for controlled structural analysis.

This framing also connects to Nissenbaum’s contextual-integrity theory, which treats privacy as appropriate information flow relative to actors, attributes, transmission principles, and social context [4]. In agentic systems, these contextual parameters become dynamic: agents may change recipients, invoke tools, write to memory, delegate subtasks, or create new derived attributes. Information-flow-control mechanisms are therefore relevant because they provide formal tools for constraining propagation across components [1]. However, adapting these mechanisms to long-horizon, LLM-mediated, multi-agent traces remains an open challenge.

12.2 Autonomy-Induced Overexposure

Early privacy mechanisms relevant to autonomous systems largely derive from access control, query privacy, and output sanitisation. Local and central differential privacy [8] provide formal guarantees for released statistics and have been adapted for telemetry and monitoring. Private information retrieval and related query privacy techniques [5, 39] reduce disclosure of query contents and access intent. Query obfuscation methods similarly aim to conceal sensitive context associated with information retrieval.

Recent work on tool-augmented and LLM-based agents highlights that autonomy changes the threat surface because agents choose what to retrieve, when to retrieve it, and where to send it, often by decomposing goals into intermediate actions that were not enumerated at design time. This weakens assumptions that access scopes and computation boundaries are static or that human oversight intervenes before each sensitive operation. Proposed mitigations include privacy-preserving interfaces, constrained tool invocation, and local adaptation with controlled noise [2, 30], but these approaches typically provide protections at specific input or output boundaries and may not compose cleanly when autonomy is distributed across multiple agents. Across this literature, overexposure is best understood as a structural consequence of self-directed action selection rather than a narrow implementation flaw, which motivates its treatment as a distinct category in the taxonomy.

12.3 Coordination and Information Propagation

Privacy risks introduced by coordination are studied extensively in distributed learning and collaborative computation. Federated learning reduces raw data centralisation, while secure aggregation and secure multi-party computation limit direct disclosure of participant updates during collaborative training or analytics. These methods provide strong guarantees for narrowly scoped collaborative functions, typically assuming synchronised execution, fixed participants, and purpose-bounded interactions.

Agentic and multi-agent settings differ in that coordination is often asynchronous, open-ended, and adaptive. Agents exchange intermediate context, partial results, and task state across time, sometimes spanning organisational roles or trust domains [30]. This creates propagation pathways that persist beyond the protected boundary of a single collaborative computation. Research on inter-agent policies and information flow control provides mechanisms for constraining message-level leakage and restricting propagation across components [1]. However, scaling such approaches to heterogeneous, cross-organisational deployments remains challenging, especially when coordination is mediated by dynamic orchestration frameworks and when policies must be enforced over both explicit messages and implicit coordination signals. These recurring observations support treating coordination-driven propagation as a distinct privacy risk category.

12.4 Memory and Persistent State Leakage

Persistent state is increasingly central to agentic systems, where memory is implemented through context windows, vector stores, caches, shared knowledge bases, and long-lived logs. Work on long-term memory in LLM agents highlights the difficulty of enforcing purpose limitation and deletion guarantees once private information is embedded, summarised, or cached for future use [27]. Related work on privacy-preserving logging and protected memory stores mitigates explicit exposure by limiting what is recorded or who can access stored artefacts [41], but it does not fully address inferential leakage arising from semantic recall and correlation across stored content.

Technical containment mechanisms such as trusted execution environments and encrypted computation can protect specific storage or execution components [19], yet their integration with dynamic and evolving agent memory architectures remains underexplored. Overall, the literature highlights a persistent gap between privacy guarantees for bounded computations and the practical governance of long-lived memory substrates in systems that continuously accumulate and reuse contextual information.

12.5 Emergent Behaviour and Compositional Leakage

Recent studies draw attention to privacy failures that arise only through composition, where individually constrained agents interact and collectively create disclosure pathways [24]. In such settings, privacy loss may occur through feedback loops, repeated interactions, and collective inference even when each component appears locally privacy-preserving. The compositional limitations of differential privacy and related mechanisms under adaptivity and correlation are well recognised, and similar degradation can occur when multiple agents generate correlated outputs over time [6].

Several lines of work attempt to model these effects using responsible agent reasoning frameworks and trace-oriented semantics, aiming to characterise privacy erosion as a behavioural property of system executions rather than as a property of a single model or dataset [2, 30]. However, many current approaches remain limited by static assumptions or lack runtime mechanisms that can adapt to evolving interaction patterns. These limitations motivate hybrid approaches that combine policy enforcement with reasoning about trace-level composition across interacting agents.

12.6 Observability and Audit Leakage

The literature on accountability and transparency identifies a tension between oversight and confidentiality. Logging, trace capture, and monitoring are essential for debugging, compliance, and governance [12], yet these artefacts often contain sensitive information by design. Studies show that unredacted logs, telemetry, and recorded intermediate reasoning outputs can expose private data to operators, auditors, or forensic analysts, even when those parties lack direct access to underlying records [30]. Proposed mitigations include privacy-preserving logging, selective disclosure, and cryptographic techniques such as zero-knowledge proofs for compliance verification. These techniques can reduce direct exposure through observability channels, but their deployment in live multi-agent systems remains limited, and they do not fully prevent inference from correlations in timing and behavioural patterns. As a result, observability mechanisms are themselves a recurring source of privacy risk in agentic and multi-agent deployments and must be treated as a first-class component of privacy analysis.

13 Conclusion

Agentic and multi-agent AI systems introduce privacy risks arising from autonomy, delegation, persistent memory, inter-agent coordination, and observability. These risks differ from those in conventional machine-learning pipelines and static distributed systems because sensitive information may be exposed through evolving execution traces, shared state, tool use, and compositional inference. This paper presented a hybrid system model that combines decentralised partially observable Markov decision processes with trace-based information-flow semantics. Based on this model, we developed a taxonomy of privacy leakage pathways, assessed the applicability of existing privacy-enhancing technologies, and identified limitations in their ability to provide adaptive and compositional protection across agents and over time. The multi-agent privacy sandbox instantiated the formal model using privacy-aware and policy-checked agents, LLM-backed tool-mediated workflows, and adversarial trace projections. The results show that layered controls reduce direct disclosure and adversarial inference advantage, while residual leakage persists through semantic cues, tool-mediated interactions, and combined trace observations. These findings indicate that established PETs are necessary but insufficient for preserving privacy in agentic and multi-agent systems. Effective protection requires their integration with privacy-aware agent architectures, runtime policy enforcement, and trace-level governance capable of regulating information flows across agents, tools, memory, and organisational boundaries. Developing mechanisms with such adaptive and compositional properties remains necessary for deploying agentic systems in environments subject to stringent privacy requirements..

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